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**मध्य प्रदेश के बालाघाट जिले में पश्चिमी उक्वा ब्लॉक के विस्तार में मैंगनीज
खनिजीकरण के लिये गुदमा ब्लॉक में प्राथमिक गवेषण
(G3- स्टेज) पर अंतिम प्रतिवेदन**

टोपोशीट क्र. 64 सी/05

(कार्यसत्र 2017-2018)

मद क्रमांक :065/एम.ई./सी.आर./एम.पी./2017/56

**FINAL REPORT ON PRELIMINARY EXPLORATION OF MANGANESE
MINERALISATION IN GUDMA BLOCK, AN EXTENSION OF WESTERN
UKWA BLOCK, BALAGHAT DISTRICT, MADHYA PRADESH
(STAGE: G -3)**

Toposheet no. 64C/05

(Field season 2017-2018)

Item code no. 065/ME/CR/MP/2017/056

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अमित कुमार, वरिष्ठ भूवैज्ञानिक

केशव खंडेलवाल, भूवैज्ञानिक

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पुर्वानुमति के बिना पूर्ण अथवा खण्डांश या किसी भी रूप में प्रकाशित अथवा उद्धरित न किया जाए प्रतिवेदन की सॉफ्ट प्रति एवं आंकिक आंकड़े महानिदेशक भारतीय भूवैज्ञानिक सर्वेक्षण, कोलकाता-700016 की अनुमति से प्राप्त किए जा सकते हैं

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**मध्य प्रदेश के बालाघाट जिले मे पश्चिमी उक्वा ब्लोक के विस्तार मे मँगनीज
खनिजीकरण के लिये गुदमा ब्लोक मे प्राथमिक गवेषण (G3- स्टेज) पर अंतिम प्रतिवेदन
(कार्यसत्र २०१७-२०१८)**

मद क्रमांक : ०६५/एम.ई./सी.आर./एम.पी./२०१७/५६

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Item code No. 065/ME/CR/MP/2017/056

(F.S. 2017-18)

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Chapter I

सारांश

भारतीय भूवैज्ञानिक सर्वेक्षण के कार्य सत्र 2017-18 में मद संख्या 65/एमई/सीआर/एमपी/2017/56 के अन्तर्गत गुदमा बालाघाट में अन्वेषण का कार्य किया गया। कार्य का मुख्य उद्देश्य मैंगनीज निक्षेप का नतिलम्ब दिशा में विस्तार देखना तथा पश्चिम उकवा ब्लॉक के विस्तार में अयस्क का अयस्क रिसोर्स बढ़ाने के लिए मूल्यांकन करना था। अन्वेषण कार्य में 1000.53 मी. वेधन किया गया तथा 44 सं. कोर नमूने, 10 सं. पेट्रोकेमिकल नमूने, 10 पेट्रोलोजिकल नमूने, 20 बी.आर.एस., 10 सं. आर.ई.ई. नमूने, 5 सं. एक्स आर एफ नमूने एकत्रित किया गया।

अन्वेषण कार्य से पता चलता है कि इस क्षेत्र में प्रीकैम्ब्रीयन आयु की शिला ओल्डर मेटामोर्फिक नाईस व सैसर मेटासेडिमेन्ट्रीस के रूप में हैं। दोनों का क्षेत्रीय नतिलम्ब ENE-WES से NE-SW है, व नति NW में मध्यम केण में है। इस क्षेत्र में विरूपण के कम से कम चार चरण हैं। पुराने मेटामोर्फिक एवं सौसर के बीच में विवर्तनीकी सम्पर्क है। इस क्षेत्र में मेग्नीज खनिजीकरण सौसर सिनसेडिमेन्ट्री-स्ट्राटाबाउण्ड एवं सौसर संघ से जुड़ा है, संभवतः लोहांगी से मंसर तक है।

वर्तमान में किये गये वेधन में अयस्क की मोटाई यह 0.24 मी. से 1.38 मी. पाई गयी है। संरचनात्मक दृष्टि से अयस्क निक्षेप किसी भी जटिलताओं से मुक्त है। अयस्क निक्षेप केवल पिंच एवं स्वैल दिखलाता है। पिंचिंग उकवा निक्षेप के दोनो साईड नतिलम्ब की दिशा के साथ ही कुछ स्थानों में नति कि दिशा में है। कलमिनेशन एवं डिप्रेषन के कारण पिंचिंग मिलती है।

गुदमा ब्लॉक में वेधन कार्य 05.11.2017 को प्रारम्भ हुआ एवं 23.02.2018 को समाप्त हुआ। कुल 1009.53 मी. वेधन किया गया जिसमें 7 बोरहोल मेग्नीज होराइजन को दो लेवल पर इंटरसेक्ट करने के लिए किये गये। अधिकतम मोटाई GDBH-1 में 1.38 मी. के साथ, 7 बोर होल ; GDBH-1ए 2,3,4,5, 6 व 7, में कुल 5.73 मी. मोटा अयस्क होराइजन मिला है एवं न्यूनतम मोटाई GDBH-4 में 0.24 मी. पाई गयी है। कुल 44 सं. कोर नमूने, 20 सं. बी.आर.एस. नमूने, 10 सं. आर ई ई नमूने, 10 सं. पी.एस. नमूने, 10 सं. पी.सी.एस. नमूने एवं 05 एक्स आर एफ नमूने एकत्रित किये गये एवं विप्लेषण किये गये।

बी आर एस नमूनों में $MnO\%$ का मान 0.03 से 30.07 % प्राप्त हुआ है। अधिकतम मान उकवा पिट व जगंटोला खदान के मेग्नीज अयस्क में प्राप्त हुआ है। पी का मान 0.03 से 3.83 % के बीच में है। तथा Fe_2O_3 0.21% से 56.21% के बीच में है।

पी सी एस नमूने नाईस, शिष्ट व फिलाईट आदि से एकत्रित किये गये। पी सी एस नमूनों में $MnO\%$ का अधिकतम मान बोरहोल GDBH-5 में सेरिसाईट शिष्ट के नमूने में प्राप्त होती है जो कि हैंगिंग वाल में है।

आर.ई.ई. नमूनों में La व Ce का अधिकतम मान 158.86 ppm व 306.89 ppm प्राप्त होता है।

जमा किये हुए 44 कोर नमूनों में $\text{MnO\%}$ का मान 0.03 से 44.94 % प्राप्त हुआ है। न्यूनतम मान शिष्ट व फ़िलाईट में प्राप्त होता है एवं अधिकतम मान मैग्नीज अयस्क में प्राप्त होता है। सिलिका का न्यूनतम मान 3.92 से 41.30 % के बीच में प्राप्त हुआ है। सिलिका का अधिकतम मान मैग्निफ़ेरस क्वार्ट्जाईट, शिष्ट व फ़िलाईट में प्राप्त होता है

क्रास सेक्सन विधि से ब्लॉक का आयतन 80849.66 मी³ एवं टनेज 168975.78 टन मिला है। तथा एल. वी. सेक्सन से ब्लॉक का आयतन 73741.84 मी³ एवं टनेज 154120.44 टन प्राप्त हुआ है। औसत Mn 21.65%, P-0.57%, Si-21.32% व Fe-4% है।



SUMMARY

The G-3 stage of exploration for manganese ore in Gudma was carried out in pursuance of FSP Item no. 065/ME/CR/MP/2017/56, of the annual field season programme of the Geological Survey of India, Central Region for the year 2017-18. The objective of the investigation was to delineate extension of Manganese mineralisation in depth and assessment of manganese ore potential in strike extension of western Ukwā block.

The exploration involved components of drilling (1009.53m), core sampling (44 nos.), petrochemical sampling (10 nos), petrological studies (10 nos.), bed rock sampling (20 nos.), REE (10 nos.) and XRF (5 nos).

The exploration work reveals that the area exposes rocks of Precambrian age in the form of older metamorphic gneisses and Sausar metasedimentaries. Both the sequences have a regional strike of ENE-WSW to NE-SW and dip at 15° to 35° angles towards NW. The area has undergone at least four phases of deformation. The contact between older metamorphics and Sausars is tectonic. The manganese mineralisation in the area is synsedimentary-stratabound and is associated with the Sausar Group of rocks, probably transgressing from Lohangi to Mansar Formation.

During current drilling, thickness of the ore horizon observed is ranging between 0.24 m to 1.38 m. Structurally the ore deposits are free from any complexities except pinch and swell structures. Pinching on the either ends of Gudma deposit is recorded along the strike. At places, along dip direction ore horizon shows pinching due to culmination and depression by interference of deformations.

Drilling in Gudma block was started on 05.11.2017 and completed on 23.02.2018. A total of 1009.53 m of drilling was done in 7 nos. of boreholes. Boreholes were drilled at two level to know strike and depth continuity of manganese horizon. The Cumulative thickness of ore horizon (ore plus waste rock) intersected in seven boreholes (GDBH-1, 2,3,4,5, 6 & 7) was 5.73m. The maximum thickness of ore horizon was observed in GDBH-1 (1.38 m) and minimum thickness in GDBH-4 (0.24 m). Core Logging of all boreholes along with geophysical logging of two boreholes GDBH-1 & 3 were done. DGPS survey was carried out to demarcate borehole points along with block boundary.

In BRS samples MnO value range from 0.03% to 30.07%, high value of MnO% 11.06 to 30.07% was found in Mn ore samples collected from Ukwa and Jagantola mine.

PCS were collected from Gneiss, Schist, Sericite schist etc. samples indicated MnO value range from 0.02% to 41.98%, high values of MnO was found in Sericite Schist samples collected from borehole GDBH-5 from hanging wall.

In REE samples La and Ce are showing higher values 158.86 ppm and 306.89 ppm respectively.

In core samples MnO% varies from 0.03 to 44.94%. Minimum value was found in Schist and Phyllite while Mn ore samples shows higher value. Minimum value of Si is varying from 3.92 to 41.30%. Higher value of silica received from Manganiferous quartzite, schist and phyllite.

By cross section method volume of block is calculated 80849.66m^3 and tonnage is 168975.78 ton and from LV section method volume of block is calculated 73741.84m^3 and tonnage is 154120.44 ton found. Average value of Mn 21.65%, P-0.57%, Si-21.32% and Fe-4% are found.

Chapter II

INTRODUCTION

II.01 FSP item no. and title

In pursuance of FSP 2017-18, vide item no. 065/ME/CR/MP/2017/56, G3 stage item was taken up by Central Region, State Unit: Madhya Pradesh, Jabalpur. The title of the project was “Preliminary Exploration of manganese mineralisation in Gudma block, an extension of Western Ukwa block, Balaghat district, Madhya Pradesh”. The exploration commenced in the month of July, 2017 (14/07/17) and was completed in the month of March, 2017 (23/03/2018).

II.02 Objectives

The objectives of project were A) To know strike extension of Manganese mineralisation in depth. B) Assessment of manganese ore potential in strike extension of Western Ukwa block.

II.03 Background Information

The main manganese producing belt in the country is associated with the Sausar Group of rocks of Pre-cambrian age. The group is characterised by lower pelitic and upper psammo-pelitic sequence. The manganese ore horizon occurs in the lower part of the Sausars.

The rocks of the Sausar Group extend in NE-SW direction in Balaghat district, M.P. in the East to West NW-ESE in Chhindwara district, M.P. in the West through Bhandara and Nagpur districts of Maharashtra, forming an overall arcuate shape with convexity towards South. The manganese bearing sausars stretches over a length of 200 km and is about 25 to 30 km wide in its central part.

The Gudma deposit is a part of Western Ukwa deposit which is located, almost, at the *Eastern* end of the belt. The Eastern part of the manganese belt, from Bharweli (Balaghat) to Ukwa. comprises, mainly, rocks belonging to the lower sequence of the Sausar Group, which is characterised by a regionally uniform strike trend of NE-SW and low to moderate dips towards NW. The ore horizon has conformable relation with the enclosing rocks and can be traced intermittently from Bharweli to Ukwa through the occurrences at Kanhatola, Laugur, Ghondi and Jagantola. (After Khal et al 1993-96).

The Ukwa manganese deposit was discovered by P.N. Bose of Geological Survey of India in 1888- 89. Geological Survey of India carried out regional exploration by means of large scale mapping, drilling, sampling etc. during 1970-71 to 76-77, and from 1993 to 1996 in lease free areas to the East and West of Ukwa deposit and also in contiguous areas of Bharweli deposit (Balaghat). Choudhary et al (1979) carried out exploration at Ukwa deposit. They drilled 41 vertical holes (GSU series) involving 5041 m of drilling and estimated gross resources of 14.71 million tonnes of manganese ore of which 12.50 million tonnes constituted the recoverable ore over a strike length of 4670 m. Devarajan and Hakim (1992) carried out Second Generation Mapping in Bharweli-Ukwa area. Their work gives detailed accounts of geology, structure and metamorphism of the basement gneisses and overlying Sausar meta-sediments. Khan and Fulmari (1996) carried out exploration work in the area in continuation of the exploration work completed earlier by Choudhary et al (1979). Large scale mapping on 1:25,000 scale has been carried out in the area between Ukwa and Baputola to have an idea about regional geology and for locating the additional manganese ore horizons if any in the dip direction of Ukwa deposit. The holes drilled by Khan and Fulmari (1996) have been marked by UKH series and except UKH 4; all boreholes are 70° South Easterly inclined.

The Gudma deposit falls under the jurisdiction of Rupjhar Police Station, Baihar tehsil, Balaghat district, M.P. It is located in the Survey of India toposheet nos. 64C/5, and is bounded by North latitudes 21°57'40", 22°58' 20" and East longitudes 80°26'20", 80°27'00".

For convenience, the entire Ukwa deposit has been divided into three blocks namely Ukwa Eastern, Central and Western blocks, as Choudhary et al (1979). In addition to above, the area between Ukwa Western block and Jagantola opencast mine has also been covered by detailed mapping and the area named as Gudma block a extension of Ukwa Western extension block.

The Eastern block extends over a strike length of 2000 m. The ore horizon crops out at around 650 m elevation at the Western end of the block and gradually rises Eastward reaching around 700 m at its Eastern end. The central block extends over a strike length of 1550 m and forms the Western continuation of Eastern block or middle part of the

Ukwa deposit. The ore horizon crops out at 630 m elevation at the Western end and rises to 650 m at its Eastern end. The Western block extends over a strike length of 1860 m and forms the Western continuation of the Central Block.

Bore holes drilled by GSI and average thickness and percentage of Mn in different blocks of Ukwa is given below:

Block	Mn % (Probable 'A' Category)	Average thickness of ore horizon	GSU series
Eastern Block	40.40	3.99 m	21
Central Block	43.14	3.51 m	21
Western Block	40.15	2.98 m	13
<i>Ukwa Deposit</i>	---	<i>3.58 m</i>	41

Except two boreholes (UKH-11 & 12) drilled at the Eastern extremity of the Ukwa deposit, all the boreholes drilled in the area by Geological Survey of India have intersected ore horizon. Thickness of the ore horizon ranges from <15 cm to about 10 m. The aggregate thickness of manganese ore in Ukwa deposit varies from 0.16 m to 4.85 m, average being 1.76 m. The aggregate thickness of waste rock (i.e. manganiferous quartzite, quartzite with rare carbonate and schist intercalation) ranges from a few cm to about 6.00 m, the average being 1.82 m. The average thickness of ore horizon in Ukwa deposit is 3.58 m, of which 1.76 m (49.16%) constitutes the manganese ore and 1.82 m (50.84%) the waste rock. Thus ore horizon generally contains 50% ore and 50% waste rock.

Ukwa deposit has been proved to extend over a strike length of 5400 m. Leaving the 480 m in long thin ore horizon at the Eastern end of the deposit, the workable strike length measures 4920 m. The ore horizon has been tested to continue in depth for more than 300 m of vertical depth in the Eastern part of the deposit and more than 200 m in rest of the deposit over major part of the strike length. In terms of dip length it measures up to 700 m from the surface exposure of ore horizon in Ukwa mine. Bottom of ore horizon has not been observed. GSI carried out extensive work in the Eastern block of Ukwa deposit by putting 30 bore holes (including GSU and UKH series). The most of the

parts of central block fall in the leasehold area. In addition to GSI, Manganese Ore (India) Limited (MOIL) also carried out exploration by means of drilling of 7 boreholes and 11 boreholes in the Eastern and central blocks respectively. Chemical and Ferro Alloy Ltd. (CFA) also drilled 11 nos. of bore holes in the central block.

In the Western block, GSI only carried out exploration (as on September 2002) and erected 24 nos. of boreholes which include GSU series and UKH series. The drilling has been carried out along section lines W-1 to W-6 placed perpendicular to the strike of the ore horizon. The spacing of the section line is 190m to 260m. The calculated recoverable resource in the Western Ukwa block is 3.01million tonnes with average Mn of 40.34%. The ore is dark grey coloured, fine grained. Ore microscopic study reveals that the Mn ore comprises braunite, hollandite, bixbyte, hausmannite, haematite, pyrolusite and cryptomelane in varying proportion. Braunite is the most abundant mineral in the ore, followed by hollandite.

The borehole interval of work carried out by Khan and Fulmari (1996) was 260 m and efforts made to get the uniform picture of the deposit upto 400m level i.e. vertical depth of 200m from the surface (600m R.L. to 400m R.L.).

During F.S.2016-17, a G-2 stage exploration item was executed in Western Ukwa block to augment the resource. The present FS programme is formulated to prove the strike extension of manganese horizon in adjacent Gudma block, SW part of Western Ukwa block. The table below indicates the mineralized horizon as intersected in the Western Ukwa block.

Depth of intersection of mineralization in Western Ukwa Block				
Section Line	BH	Total Drill Depth in meter	Mineralized Zone (MZ) in meter	Thickness of MZ in meter
AB	BH-06 (UKBH-1)	113.35 m	82.5 -83m	0.5m
	BH-08 (UKBH-5)	200 m	197-198.8m	1.8 m
CD	BH-05 (UKBH-2)	116 m	100.3-102.35	2.05 m
	BH-07 (UKBH-4)	254 m	251.8-253.5 m	1.70 m
EF	BH-04 (UKBH-3)	128 m	123.7-127.31	3.61 m
	BH-09 (UKBH-6)	260 m	255.5-256.3	0.8 m

During FS 2016-17, it was found that manganese ore horizon is still continuing in strike direction as supported by MOIL open cast mine. With this background, it was proposed for 12 bore holes in Gudma block, in strike continuation of Western block of the U kwa deposit for a total of 1000m drilling around 130m borehole interval during FS 2017-18 along with detail mapping on 1:2000 scale. Bore holes were planned to intersect ore horizon by putting 12 bore holes that is 6 bore holes were planned to intersect the ore body at first level (around 50m vertical depth) and 6 boreholes to intersect the ore body at second level (around 100m vertical depth). However due to ground conditions 5 boreholes were drilled to intersect the ore horizon at second level and 2 boreholes were drilled to intersect the ore horizon at third level.

II.04 Nature and Quantum of work

Physical target achieved during FS 2017-18 along with field stay details and expenditure details are provided below in tabular form.

Table-I Quantum of work done vis-à-vis achievement for FS: 2017-18

Nature of work	Total work load envisaged	Total
1. Technological		
(a) DM (1:2000)	0.5 sq km	0.5sq.km
(b) Subsurface exploration		
i) Drilling (In-house unit-414 & 416)	1000m	1009.7m
i) Core sample	44 nos.	44*
2. Geochemical Survey		
(a) Surface Exploration		
(i) BRS	20 Nos.	20
3. Petrographic/Mineragraphic Studies		
a) PS	10 Nos.	10
a) PCS	10 Nos.	10
a) XRF	5 Nos.	5

***Including 7nos of check samples**

Table-II Field stay and RCA expenditure

Name of Officers	Field Days
1. Amit Kumar, (Sr. Geologist)	119
2. Keshav Khandelwal,(Geologist)	115
Supervisory Officer A.K.Talwar, Director.	10

Total Expenditure in Rs.
2,64,021/- (Wages)
56,643/- (POL)
30,896/- (OC)
3,51,560/- Total

II.05 Acknowledgement:

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CHAPTER III

PROPERTY DESCRIPTION

III.01 Title of Ownership

Geological Survey of India, Central Region, State Unit-Madhya Pradesh, Jabalpur carried out “Preliminary Exploration of manganese mineralisation in Gudma block, an extension of Western Ukwa block, Balaghat district, Madhya Pradesh (Stage G-3)” project under Mission-II, Ferrous Minerals, Mineral Resource Assessment for the Field Season 2017-18 vide FSP Item Nos. 065/ME/CR/MP/2017/56.

III.02 Details of the area

III.02.1 Village name and district

The current study area belongs to Gudma village of Baihar Tehsil, Balaghat District, State M.P.

III.02.2 Survey of India Toposheet No.

The current study area falls under Survey of India Toposheet no. 64C/05.

III.02.3 DGPS Geo-coordinates.

DGPS co-ordinates of the block marked on plate-VI having current boreholes are as follows:

A 21° 57' 53.299"N	80° 26' 20.364" E
B 21° 58' 8.832" N	80° 26' 39.944" E
C 21° 58' 1.210" N	80° 26' 46.255" E
D 21° 57' 46.105" N	80° 26' 26.624" E

The co-ordinates are written in counter-clockwise manner.

III.02.4 Cadastral details

The present block of drilling is a part of Gudma block, an extension of Western Ukwa block, which comes under the revenue land. Southern area from Gudma block is allocated to PSU (MOIL). The area from June to October was covered by paddy fields for this reason the drilling work was initiated in month of November. Prior permission from land owner was taken by drilling unit in charge associated in project. As first series of

boreholes falls under MOIL leasehold boundary therefore borehole proposal was modified after field check. A total of 5 nos. of boreholes of second level and 2 no of borehole of third level were drilled.

III.02.5 Freehold/ Leasehold

The present drilling was performed in freehold area, however prior permission of private land owner was taken by drilling unit in charge associated with project.

III.02.6 Location and accessibility

The present study area falls in T.S. Nos. 64C/05 and bounded by North latitudes 21°57'40.90", 21°58' 08.26" and East longitudes 80°26'13.21", 80°26'58.07". The area of investigation falls under the jurisdiction of Rupjhar Police Station, Baihar tehsil, Balaghat District, MP. Gudma is situated about 2.5km from U kwa town. The U kwa mine and town are located 42 km ENE of Balaghat, on Balaghat-Baihar road (SH-26). The nearest railhead is Balaghat junction located on Jabalpur-Gondia broad gauge line of the South Eastern Railways for which construction is still in progress. The interior parts of the area on the either sides of the State Highway are connected with the network of all weather metalled road and fair weather kuchcha roads.

III.02.7 Climate

The area experiences varied climate. Summer commences from the end of March and continues up to middle of June, with May being the hottest month. During summer day temperature rises up to 45°C, but nights are pleasant. Winter is pretty cold with some times temperature dropping down to 3°C. Frosty conditions prevail during winter nights. Intermittent rains in winter are very common. December and January are the coldest months. Area receives rainfall from southwest monsoon with the July and August being the months experiencing maximum rains. The average annual rainfall is around 1537 mm.

III.02.8 Flora and fauna

The area has forests of dry, mixed deciduous type with bamboo. Teak and Sal plantations are scattered. Some of the important flora found in the forests are Saja (*Terminalia tomentosa*), Dhaman (*Grewia tiliaefolia*), Teak (*Tectano grandis*), Sal (*Shorea robusta*), Bija (*Steroosperma suavele nnis*), Dhure (*Anoaeissue latifolia*), Kumbhi

(Gareva arborea), Tinsa (*Quoeina calberolodes*), Harra (*Terminalia chebula*), Behara (*Terminalia belirica*), Jamun (*Eugenia jambolana*), Tendu (*Diosplyros mclanoxyla*), Mahuwa (*Madhuka latifolia*) and Lendia (*Lagerstromia paulifora*).

The common wild animals found in the region are tigers, panthers, wild cats, hyena, wild dogs, bisons, sambars, wild boars, bears, barking deer and red and black-faced monkeys. The area around Ukwa is full of Cobras. Apart from the wild animals and reptiles, the area has various types of birds. The Kanha National wild life sanctuary is around 45 km northeast of Ukwa.

III.03 Infrastructure & Environment

The social infra structure in and around Gudma area includes: Primary health center, Ukwa; Government Primary School, Jagantola; and Communication & Post-office, Rupjhar. Kanha National Park is world famous national park and attracts lots of tourist for its beauty and presence of maximum numbers of tiger there. Topographically the area is located on the Baihar plateau and is drained by the tributaries of Wainganga River, namely, Nahra Nadi and its off shoots. The area around Gudma is almost plain with very low south westerly gradient. The area between this plain and flat-topped Tipagarh hill range is quite rugged. The manganiferous horizon occurs as linear ridge striking ENE-WSW to NE-SW within the plain country of Ukwa, starting from Tipagarh and continuing southwestward up to Jagantola. The highest point is 800 m at the fringe of Tipagarh range (NE corner of toposheet No. 64C/5) and lowest around 570 m along Nahra Nadi south of Dora. In ground conditions at present ridge of Mn has been replaced by open cast mines of MOIL. Ground water condition is quite good. The use of ground water is almost negligible in the area. Mica schist being the major litho unit in the country acts as good aquifer. Luxuriant growth of natural vegetation like palm and jamun trees indicates the shallow water table condition.

CHAPTER IV

PREVIOUS EXPLORATION

IV.1 Details of previous exploration

Bose P.N. (1888-89) of Geological Survey of India discovered the Ukwā manganese deposit. Fermor L.L. (1903-04) studied the important manganese deposits of the region and proposed the name 'Gondite series' for the manganiferous suite of rocks in Chhindwara, Nagpur and Balaghat areas. Burton R.C. (1911-12) carried out Geological surveys in Balaghat district. Shukla K.D. alone in 1950-51, and Shukla K.D. and Anandalwar M.A. in collaboration with Stracjek J. A. of USGS carried out geological mapping in the area and paid special attention to the manganese ore deposits of the region (1,2,3,4,5 source - Shukla and Anandalwar, 1973).

MOIL (the then Central Provinces Manganese Ore Ltd.) carried out core drilling in Ukwā deposit at 19 sites during 1955-60. Geological Survey of India carried out regional exploration by means of large scale mapping, drilling, sampling etc. during 1970-71 to 76-77, in leasefree areas to the east and west of Ukwā deposit and also in contiguous areas of Bharweli deposit (Balaghat). The areas covered under the regional exploration are as follows:

1970-71 Jagantola-Ghondi, Dhamsua-Laigur and Laigur-Kanhatola sectors.

1971-72 Ukwā-Gudma block.

1972-73 Jagantola-Ghondi, Dhamsua-Laigur sectors and Balaghat-Aolajhiri block.

1974-75 Jagantola-Ghondi sector and NE extension Samnapur block.

1975-76 Dhamsua-Laigur. Laigur-Kanhatola sectors and Balaghat-Aolajhiri block.

1976-77 Dhamsua-Laigur and Laigur-Kanhatola sectors. Results of the above explorations are covered under unpublished reports of Geological Survey of India (Source -Choudhary et al; 1979).

Exploration by regional drilling for manganese ores in Ukwā-Gudma Block (included in above list) was carried out by Akolkar (1974). Based on the results of two regional boreholes, he indicated possible reserves of the order of 0.525 million tonnes of low grade, medium to high phosphorus ore, of which 0.263 million tonnes constituted as recoverable one.

Exploration by drilling in Jagantola-Ghondi and northeast extension of Samnapur sectors, forming SW and NE extension of Ukwa deposit respectively were carried out by Bhoskar and Rao (1975). In Jagantola-Ghondi sector, they indicated the aggregate 'insitu' and inferred reserves of 2.08 million tonnes, of which 0.86 million tonnes constituted as recoverable ore containing 30 to 38 percent manganese. In Samnapur extension sector the manganese ore was more or less gonditic.

Ukwa deposit was also explored by Choudhary et al. (1979). They drilled 41 vertical boreholes involving 5041 m of drilling and estimated gross reserves of 14.71 million tonnes of manganese ore of which 12.50 million tonnes constituted the recoverable ore over a strike length of 4670 m.

Pal and Deshmukh (1974) studied the thin siltstone bands associated with the manganese ore horizon. They concluded that the siltstones were of sedimentary origin and were probably present in metamorphic shadow zone.

Devarajan and Hakim (1992) carried out Second Generation Mapping in Bharweli-Ukwa area. Their work gives detailed accounts of geology, structure and metamorphism of the basement gneisses and overlying Sausar metasediments.

Khan and Fulmari (1996) carried out exploration and drilling in continuation of the exploration work completed earlier by Choudhary et al. (1979) and for deeper level drilling the section lines used by Choudhary et al; was extended further northwestward by them. They completed 4574.22 m of drilling and carried out 51 sq km large scale mapping on 1: 25,000 scale and 5.18 sq km detail mapping on 1:2000 scale. Large scale mapping on 1:25,000 scale has been carried out in the area between Ukwa and Baputola to have an idea about regional geology and for locating the additional manganese ore horizons if any in the dip direction of Ukwa deposit.

Detailed mapping (1:2000) of entire Ukwa open cast mine, adjoining foot wall and hanging wall rocks were covered prior to drilling and for convenience the entire area has been divided into three blocks namely Ukwa Eastern, Central and Western blocks. They have also reported that the average Mn content of the Ukwa deposit is 41.23% and P is 0.23%. Major part of the deposit falls under medium grade ore having 35 to 45% Mn. Phosphorus content increases in Western block and also at depth in other blocks. The

deposit has estimated gross ore reserves of 16.23 million tonnes which would involve mining of 29.07 million tonnes of material consisting of ore and waste rock i.e. the run of mine. The recoverable reserves (90% of gross reserves) come to 14.61 million tonnes representing 50.26% of run of mine, of which 11.84 million tonnes fall under Probable 'A', 1.37 million tonnes under 'Probable' and 1.40 million tonnes under 'Possible' category. The favourable geological and structural set up of the ore horizon may lead to easy minability of ore, but the high phosphorous content at depth could be a discouraging factor.

Kumar and Khandelwal (2016-17) carried out general exploration for manganese mineralization in western block of Ukwa, Balaghat district, Madhya Pradesh. The exploration involved drilling of 1071.35 m, core sampling of 51 nos, petrochemical sampling of 10 nos, petrological studies of 10 samples, bed rock sampling of 20 nos. REE 10 nos. and XRF 5 nos. For resource estimation; earlier worker data was combined with current data to find out the weighted average of current block and it is found to be 35.138%. Based on this grade specific gravity as described by previous workers on this range is 3.55 (90% of actual value considering 10% loss) is used and resource estimated is thus 2.182 mt by cross section method and 2.013mt by LV section method at 35.138% of Mn. Here resource estimation of current area is also determined by previous workers sections using their LV section and previous chemical values; based on which resource was found to be 3.597 mt at 40.34% of Mn.

IV.2 Details of Geophysical Investigation

Gaonkar S.G. and Das A.K. carried out geophysical surveys utilizing gravity, magnetic (V.F.) and self potential methods in Ukwa (21°58'30", 80°28'W) area, Balaghat district, M.P. for finding out extension along strike for the Ukwa manganese deposit in the soil covered gap area between Ukwa deposit and Jagantola trench in the Western Block and to the east of the Eastern Block.

The strata bound manganese bodies are located between the mica schist and the phyllites metasedimentaries. The metasedimentaries are overlying the granite gneiss basement. The bedding and the foliation trends are along the ENE-WSW direction with

moderate dips of about 25° to NW. The manganese horizon is exposed over a strike length of about 5 km and the thickness of the horizon ranges from 0.27 m to 6.1 m.

Two major magnetic anomalies, in the form of linear bands with strike length of about 3.5 km each and magnitude in the range of 100 to 400 gammas were picked up in the western block of this area. The width is about 25 and 100 m for the southern and the northern anomalies respectively. The southern magnetic high was recorded over the main Ukwa mine trench, where the ore body with average width of 3 m is exposed at the surface. The anomaly shows a dominant positive high with absence of negative low to the north. This shape is attributed to the northward dip of the ore body. The gravity surveys closely corroborate the results of magnetic surveys in this area. Two linear gravity anomalies corroborate with the observed linear magnetic anomalies. The geophysical surveys indicate strike continuation of the exposed manganese mineralisation under soil cover, from Ukwa trench in the east to Jagantola trench in the west, for over 1.6 km. Presence of another gravity-magnetic linear anomaly in the soil covered area to the north of the ore body exposed in the main trench indicates interesting possibilities for the presence of manganese mineralisation. S.P. surveys were also carried out in this area. The surveys have brought out feeble but definite low values of the order of 20 to 80 mv and corroborate with the gravity magnetic response over the manganese ore body exposed in trench and also support its extension to the west over a strike length of 1.6 km. No S.P. response was recorded over the northern gravity magnetic anomaly band. This could be due to the thick soil cover and the causative body being at greater depth.

A magnetic peak with narrow width was picked up over manganese pit on Tr. TU4 in the eastern block. However, this anomaly was not found to continue over the next traverse located at an interval of 300 m to east. From the appearance of rapid fluctuations in magnetic values on traverses laid further to the east in this block, it is inferred that Sausar formations have gone under Deccan Trap rubble.

As per geophysical investigation, drilling was recommended in the gap area between Ukwa and Jagantola trench in western block to test the possibility of manganese mineralisation in this area.

CHAPTER V

GEOSCIENCE INVESTIGATION

V.01 Regional Geology

The main manganese producing belt in the country is associated with the Sausar Group of rocks of Pre-cambrian age. The group is characterized by lower pelitic and upper psammo-pelitic sequence. The manganese ore horizon occurs in the lower part of the Sausars.

The rocks of the Sausar Group extend in northeast-southwest direction in Balaghat district, MP., in the East to West NW-ESE in Chhindwara District, in the West through Bhandara and Nagpur districts of Maharashtra, forming an overall arcuate shape with convexity towards South. The manganese bearing sausars stretches over a length of 200 km and is about 25 to 30 km wide in its central part. The Ukwa manganese deposit is located, almost, at the eastern end of the belt.

The eastern part of the manganese belt, from Bharweli (Balaghat) to (Gudma) Ukwa, comprises, mainly, rocks belonging to the lower sequence of the Sausar Group, which is characterized by a regionally uniform strike trend of NE-SW and low to moderate dips towards NW. The ore horizon has conformable relation with the enclosing rocks and can be traced intermittently from Bharweli to Ukwa through the occurrences at Kanhatola, Laugur, Ghondi and Jagantola.

The low-lying areas around Ukwa, bordering the plateau, predominantly comprise rocks of Sausar Group. The Sausar rocks occur in the areas from southern vicinity of the existing manganese mine pits to more than 15 km north of Ukwa mines. The areas South of mine pits expose older metamorphics in the form of gneisses and associated rocks. A solitary thin bed of pale gritty clay belonging to Lameta beds, unconformably overlying the Mansar schist of Sausar group is recorded in Ghondi mine section. The Deccan Traps occur at higher levels forming plateau. The contact between Sausars and Deccan traps occurs around 700 m above m.s.l. The traps are capped by about 25 m thick laterite. Stray laterite mounds are also seen over older metamorphics and Sausars. The laterite contains pockets and lenses of bauxite.

The generalized succession established by Shukla and Anandalwar (1973) around U kwa region is presented in below Table III.

Table: III - Generalised geological succession of U kwa area (after Shukla and Anandalwar, 1973).

Pre-Cambrian Group	Formation	Brief description of rock types
	Recent and Sub - Recent	Sandy soils and alluvium
	Tertiary	Laterite and laterite gravels
	Post Sausar Intrusives	Pegmatite and vein quartz, Granite (?) Basic dykes and orthogneisses
	Bichua Formation	Marbles, Calc silicate granulite
	Junewani Formation	Quartz-biotite schist with garnet, Kyanite, staurolite
	Chorbaoli Formation	Quartz-biotite schist with garnet and Kyanite, micaceous-flaggy quartzites, quartz schist
	Mansar Formation	Muscovite schist, quartz-muscovite schist, quartz-sericite-muscovite schist with garnet
	Lohangi Formation	Pink calcitic marbles (impersistant)
	Sitasaongi Formation	Quartzite, quartz-muscovite schist and conglomerate. Felspathic grits and conglomerate
Archaeon (Unclassified)	Older metamorphics	Ortho and paragneisses, composite gneisses and granite, bands and lenses of amphibolite, epidiorite, quartzite etc, with pegmatites and quartz veins

Only a part of the above succession is exposed in the area around Gudma where exploration work has been carried out. The older metamorphics are exposed in the area South and SE of U kwa mine and outcrop can be seen near Samnapur Nala. The major part of the area is occupied by Mansar Formation and perhaps the litho unit exposed around Nagjhola and East of Khursurtola in Nalira Nadi belongs to Chorbaoli Formation.

All the litho units have ENE-WSW regional trend with low to moderate NW dip around Ukwa , Gudma, but there is a general tendency of steepening as one goes further North.

V.01.1 Geological set up

Khan and Fulmari, 1996 carried out exploration an area of 51 sq.km and mapped around Ukwa on 1:25,000 scale with a view to have better understanding of the geological set up and to find out additional ore horizon, if any. The area covered by large scale mapping exposes the older metamorphics and mainly pelitic schists with manganese ore horizon. The succession examined in the area is indicated in below *table IV*.

Table: IV – Geological Succession in Ukwa-Baputola area (Khan and Fulmari 1996).

Pre-Cambrian Sausar Group	Recent to sub-Recent	Soil and alluvium
	Tertiary	Laterite
	Post Sausar	Pegmatite and vein quartz
	Chorbaoli Formation (?)	Garnetiferous quartz-muscovite-biotite schist $\pm$ Kyanite
	Mansar Formation	Garnetiferous muscovite schist
		Garnetiferous sericite schist/phyllite
		Garnetiferous quartz muscovite schist with intercalations of thin quartzite
		Sericite schist. Manganese ore horizon, Interbanded Mn ore and manganiferous quartzite.
	Lohangi Formation (?)	Phyllite with calcareous bands
Tectonic contact		
Archaean	Older metamorphics	Granite gneiss with enclaves of biotite schist, amphibolite, intruded by K-granite, pegmatite and vein quartz

V.02 Detailed geological exploration

Khan and Fulmari, 1996 carried out detailed mapping of 5.18 sq.km area in and around Ukwa on 1:2,000 scale. The area covered by detailed mapping exposes ENE-WSW trending ridges and mostly paddy fields.

V.02.1 Description of Rock types

The area can be divided in two part viz, a) older metamorphic and b) Sausar group. Older Metamorphis represented by Gneiss, enclave of schist and amphibolite while the Sausar group is represented by schist and phyllite with Mn ore.

V.02.1.1 Older Metamorphics

The older metamorphics occur in the area to the immediate South of the mine pit and they continue down South beyond the area of exploration. The unit is represented by ortho and para gneisses which contain enclaves of amphibolite and biotite schist (*Figure V.1A,B*). The amphibolite enclaves are rare in the area adjacent to the Ukwa mine. Their intensity increases further South. These enclaves are more susceptible to lateritisation. The laterite cappings occurring in the area South of Samnapur are perhaps developed on these enclaves. Biotite schist occurs as thin bands and lenses up to 50 cm in thickness within the gneisses. The rock is well foliated, and bands are stretched along the regional foliation of the gneiss.

The gneisses are banded and augen type composed mainly of quartz, plagioclase, K-feldspar and biotite with subordinate muscovite and chlorite. At places gradation is observed between gneisses and the biotite schist, amphibolite and biotite granite, from which they have been derived (After Khan et al 1993-96). The gneiss is exposed in Bharveli – Ukwa road section near Gangulpara area (*Figure V.2A,B*).

The gneisses are intruded by fine grained pink granite, aplite, pegmatite and vein quartz. These intrusives are concentrated along the contact between gneisses and overlying metasedimentaries. The pegmatite and vein quartz are seen in both the gneisses and metasedimentaries, but no traces of granite and aplite have been recorded in metasedimentaries on the surface or in any boreholes drilled by Geological Survey of India. Hence, these granite and aplite are either part of the older metamorphics or have been emplaced just at the beginning of Sausar orogeny, as they have been involved in the tectonic movement taken place between the gneisses and metasedimentaries (After Khan et al 1993-96). The gneisses and granite both are, at places, quite rich in magnetite. Magnetite occurs as fine disseminations or mm thin stringers along the fractures. Pyrite also, occurs in the similar fashion either in association with magnetite or without it. In general gneisses are light grey in colour but wherever they are intruded by potassic granite they show pinkish grey to pink colour. This granite exposed near Lud area on Ukwa to Bithli road section (*Figure V.3B*).

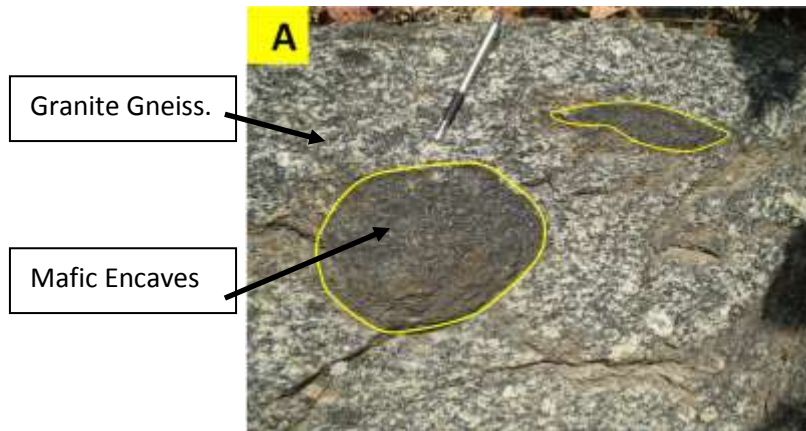


Figure V.1A : Older metamorphic enclaves(schist)exposed on road section near Gangulpara. T.S. no. 64C/05



Figure V.1B : Older metamorphic enclaves exposed on road section near Laugur. T.S. no. 64C/05



Figure V.2A : Biotite Gneiss exposed on Samnapur Nala Section in T.S. no. 64C/05



Figure V.2B: Migmatite Gneiss exposed on Bharveli-Ukwa road Section in T.S. no. 64C/05



Figure V.3A : Porphyritic granite exposed on Bharveli-Ukwa road Section in T.S. no. 64C/05



Figure V.3B : Granite exposed near Lud area in T.S. no. 64C/05

V.02.1.2 Phyllite/ Schistose rock

Metasedimentaries, the lower most unit occurring in the area is phyllite with calcareous intercalations. The phyllite is chloritic and sericitic compositionally. At places it grades into sericite schist. The calcareous intercalations in this unit are represented by quartz-sericite/muscovite-calcite schist.

The presences of calcareous intercalations indicate that this unit belongs to Lohangi Formation. This unit is quite persistent forming foot wall rock for manganese ore horizon as evident during drilling. Occasionally sericitic quartzite interbands are also noticed in this unit. The calcareous intercalations are commonly recorded in footwall rock, but very often they are also noticed interbanded with the lower part of ore horizon, thus, perhaps, indicating the gradational relationship between Lohangi and Mansar

Formations. The general trend of the units is N60°E-S60°W with 25° to 35° dip towards NW (*Figure V.4A and B*).

The uppermost unit of Mansar Formation is Garnetiferous Quartz muscovite schist. This unit occurs in hanging wall above Mn ore horizon. This lithounit is well foliated and fine to medium grained. Major mineral are muscovite, Quartz and Garnet. The general trend of this unit is vary from N45°E-S45°W to N60°E-S60°W with 25° to 35° dip towards NW.

Petrographic study of schist reveals that the rock represents a garnet-quartz-muscovite-biotite- schist with accessory opaques, tourmaline, apatite and monazite. The rock is compositionally banded with alternate layers rich in micas and quartz. Calcite is more concentrated in the coarser quartz-rich layers in calc-schist while Qtz is dominating in Qtz-Muscovite schist. Garnet crystal is embedded in a muscovite-biotite. Under microscope it showing well developed scistosity and crenulation. (*Figure V.5A,B*). Refraction cleavage can be identified easily in thin section (*Figure V.10B*).



Figure V.4A : Garnetiferous Quartz Muscovite schist exposed on Bharveli-Ukwa road Section in T.S. no. 64C/05



Figure V.4B : Manganese dendrite in Sericite schist exposed on Jagantola Mine in T.S. no. 64C/05

V.02.1.3 Manganese ore horizon

Mn ore occurs as bands interbanded with cherty manganiferous and ferruginous quartzite and impure calcareous rock as quartz-sericite/muscovite-calcite schist. Apart

from manganiferous and ferruginous quartzite and calcareous rock, Mn ore is also interbanded occasionally with sericitic phyllite/schist, particularly at the lower portion of the horizon. The individual bands of quartzite and calcareous rock varies in thickness from <1 cm to 30 cm. The ore is either massive in nature or is banded with mm thin quartzite and carbonate partings. The ore is dark grey coloured and fine grained. The Mn ore horizon is continuous band (*Figure V.6A and B*) showing culmination and depression nature and due to deformation in the area. It occurs as massive, high grade ore as well as interbanded with quartzite, low grade ore. Total thickness of Mn ore intersected in 7 different boreholes (GDBH-1, 2, 3, 4, 5, 6 & 7) is 4.36m. The maximum thickness of ore horizon was observed in GDBH-1 (1.38m) and minimum thickness in GDBH-4 (0.25m).

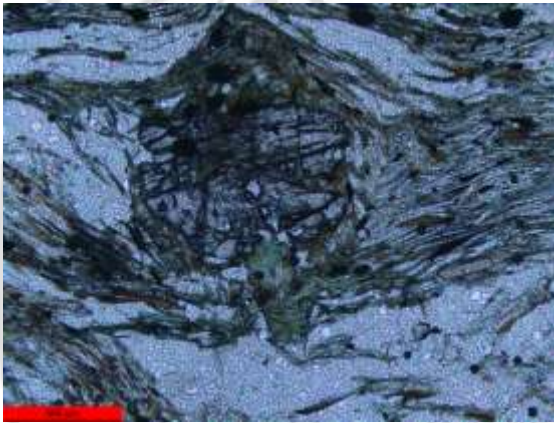


Figure V.5 A : Photomicrograph shows Garnet Porphyroblasts embedded in a muscovite-biotite under PPL

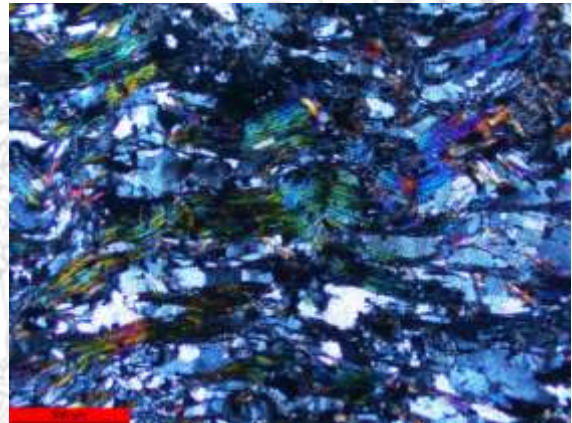


Figure V.5 B : Photomicrograph shows schistosity with crenulations in Garnetiferous quartz muscovite schist under XPL

V.02.2 Structures

V.02.2.1 Primary Structures

In metasedimentaries the most important non diastrophic structure is bedding plane (So) (*Figure V.6A and B*), which is very well preserved in ore horizon in the form of compositional layering represented by alternations of Mn ore and manganiferous quartzite, with occasional calcareous and rare argillaceous (sericitic schist) bands. The colour/compositional layering in footwall phyllite also represent the bedding plane of the

rock. The banded nature of quartz-mica schist and calcareous mica schist is also because of original compositional layering. The bedding in ore horizon and associated footwall and hanging wall rocks strikes NE-SW to ENE-WSW and dips at low to moderate angles towards NW.



Figure V.6A: S-0 as observed in Manganese ore horizon in MOIL mine pit, Ukwa (B), T.S. no. 64C/05.

Figure V.6B: Undulations in Manganese ore horizon due to deformation in MOIL mine pit, Ukwa (B), T.S. no. 64C/05.

V.02.2.2 Secondary Structures

Secondary structures in the gneiss are in the form of gneissic foliation and mineral lineations. The secondary structures in the metasedimentaries are in the form of axial plane schistosity (S_1) of regional (F_1) folds with which is associated intersection lineation (L_1) formed by the intersection of bedding with schistosity (*Figure V.10A, B*). The axial plane cleavage (S_2) of second-generation folds (F_2) is associated with minor folds and intersection lineation (L_2) formed by the intersection of axial plane with S_0 or S_1 . In the similar fashion the evidences of F_3 and F_4 and related planar and linear structures i.e. S_3 , L_3 and S_4 and L_4 are also recorded in the area which are better observed and understood in metasedimentaries. (After Khan et al. 1993-96)

The structural data reveals that the area has undergone at least four phases of folding deformation (D_1 to D_4 / F_1 to F_4). The first phase (D_1) was the major one which

gave rise to ENE-WSW major (F_1) overturned folds. The Sausars present in Ukwa area are considered to be the southeastern limb of an overturned synform and regional foliation (S_1) as axial planar to F_1 folds. This phase has given rise to subhorizontal intersection lineations (L_1) formed by intersection of S_1 on S_0 plane. The second phase (D_2) has been represented by low west to WSW plunging minor folds (F_2) which has developed subvertical crenulations (S_2) on S_0 and S_1 planes and L_2 intersection lineations and fold axes. (After Khan et al; 1993-1996.).

The third phase (D_3) is not very well developed throughout but wherever recognizable, it is seen in the form of NNE-SSW trending warps (F_3) and associated axial planar cleavage (S_3) and intersections lineation (L_3) and fold axes (*Figure V.8A,B*). The S_3 strikes in NNE-SSW direction and dips sub vertically. L_3 is low NNE plunging (After Khan et al; 1993-1996.)

The fourth phase of deformation (D_4) has given rise to NW-SE trending open warps on earlier formed planar structures, producing depressions and culminations. The F_4 are low north-westerly plunging with subvertical axial plane giving rise to NW-SE striking and sub vertical dipping axial plane cleavage (S_4) and low to moderate NW plunging intersection lineations (L_4) and fold axes (*Figure V.9A*) (After Khan et al; 1993-1996.)

The culminations and depressions in the levels of intersection of ore horizon is perhaps because of the combined effects of the different phases of deformations).

The contact between older metamorphics and Sausars is tectonic which is also evidenced by mylonitization of gneisses along the contact. This feature is perhaps related to the first phase of deformation (D_1). It indicates that D_1 is represented by F_1 folding and probably thrusting also. The event has imparted shear foliation on gneisses showing parallelism with the S_1 of Sausars. Perhaps the original planar structure (gneissosity) of older metamorphics has been obliterated by D_1 along the contact with Sausars. (After Khan et al. 1993-1996)



Figure V.7 : M type fold present in granite gneiss exposed North-West of Gudama block, Ukwu.

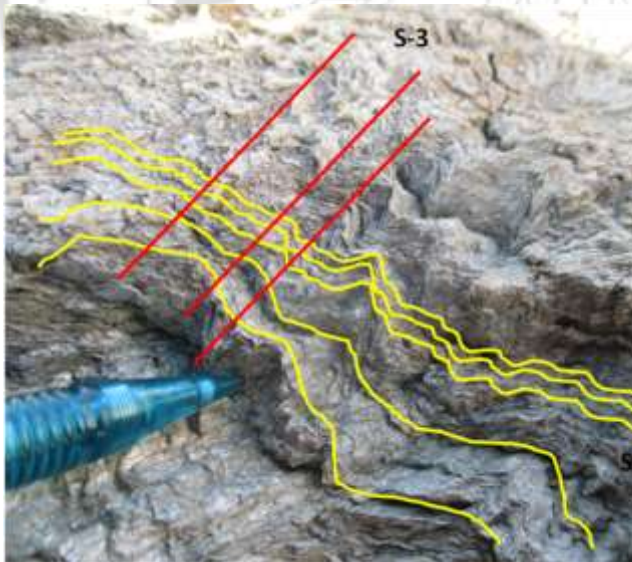


Figure V.8A : Sericite schist undergone different phases of deformation



Figure V.8B : Phyllite showing S-3 plane as observed in near Banjari, T.S. no. 64C/05 in cross section dipping NW.



Figure V.9 : Phyllite showing S-4 plan as observed in near Banjari, T.S. no. 64C/05 in cross section dipping NW.

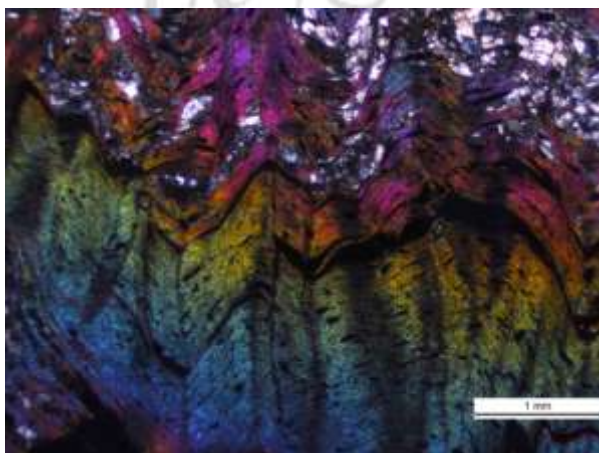


Figure V.10A : Photomicrograph shows crenulations in S1 in Quartz-muscovite-biotite schist.

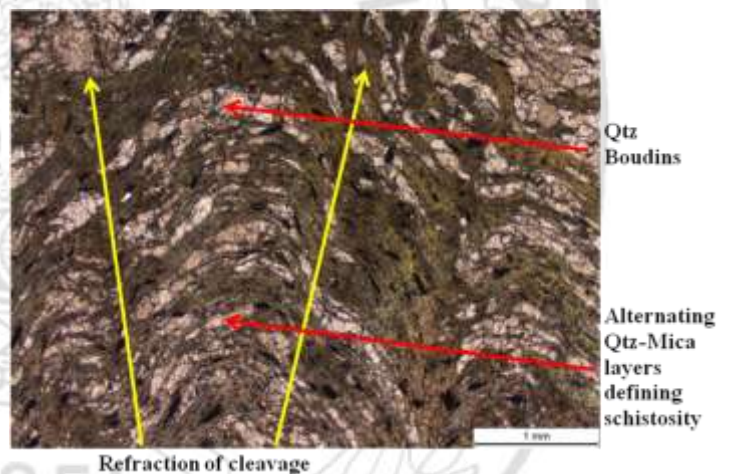


Figure V.10 B : Photomicrograph shows Quartz-Muscovite Schist indicating schistosity and cleavage refraction

V.02.3. Metamorphism

The area have undergone metamorphism of lower granulite facies in the Bharweli Ukwa area. The Sausar have undergone amphibolite facies of metamorphism with the successively higher zones being encountered from south to north. The low grade phyllites

of the Mansar Formation partly belong to the greenschist facies. The chief constituent minerals are muscovite, chlorite and quartz. The associated pelitic schists of Mansars have undergone metamorphism of chlorite-biotite zone. The major constituent minerals are quartz and muscovite with subordinate amount of biotite and chlorite. (Figure V.5AB and V.10AB).

V.02.4 Mineralogy of the ore zone and ore texture

Mn ore comprises braunite, hollandite, bixbyite, hausmannite, haematite, pyrolusite and cryptomelane in varying proportion. Braunite is the most abundant mineral in the ore, followed by hollandite. Texturally the ore is either massive or banded. The massive type comprises mostly braunite and hollandite with granoblastic fabric while the banded ore has thin but distinct bands rich in braunite or hollandite or bixbyite.

By the ore beneficiation study it was found that in area major amount of braunite and quartz with subordinate amount of psilomelene, cryptomelene and carbonate (rhodochrosite, dolomite and calcite), minor amount of garnet and pyrolucite and very minor amount of feldspar (plagioclase, microcline), jacobsonite, amphibole, mica, pyroxene, hausmanite & rhodonite. Hollandite, tourmaline, clay, epidote, hematite and goethite/limonite is present.

V.02.5 Pitting and trenching

No pitting and trenching was carried because in work area ore horizon lies very deep.

V.02.6 Sampling

V.02.6.1 Core Samples

Core samples of -120 mesh size were prepared and submitted after drilling was completed. Details of sampling methodology is explained in chapter IX and weighted average calculation is attached in Annexure-VII . A total number of 44 Core samples was prepared including check samples. Details of samples collected from individual boreholes are described below (table V):

Table-V Details of core samples

Borehole No	From	To	Sample No	Lithology	Mn%
GDBH-1	99.57	99.82	CS/01/ JBP/GDM/ 2017-18/GDBH-1	Sericite Schist	0.05

GDBH-1	99.82	100.269	CS/02/ JBP/GDM/ 2017-18/GDBH-1	Mn Ore	32.64
GDBH-1	100.269	100.575	CS/03/ JBP/GDM/ 2017-18/GDBH-1	Mn Ore	41.18
GDBH-1	100.575	100.9	CS/04/ JBP/GDM/ 2017-18/GDBH-1	Mn Ore	44.95
GDBH-1	99.57	101.2	CS/05/ JBP/GDM/ 2017-18/GDBH-1	Manganiferous quartzite	5.65
GDBH-1	100.9	101.2	CS/06/ JBP/GDM/ 2017-18/GDBH-1	Mn Ore	34.04
GDBH-1	101.2	101.45	CS/07/ JBP/GDM/ 2017-18/GDBH-1	Phyllite	0.77
GDBH-2	97.2	97.45	CS/01/ JBP/GDM/ 2017-18/GDBH-2	Sericite Schist	0.08
GDBH-2	97.45	97.77	CS/02/ JBP/GDM/ 2017-18/GDBH-2	Mn Ore	35.09
GDBH-2	97.77	97.94	CS/03/ JBP/GDM/ 2017-18/GDBH-2	Mn Ore	41.13
GDBH-2	97.45	98	CS/04/ JBP/GDM/ 2017-18/GDBH-2	Manganiferous quartzite	14.66
GDBH-2	98	98.25	CS/05/ JBP/GDM/ 2017-18/GDBH-2	Phyllite	0.25
GDBH-3	146.65	146.91	CS/01/ JBP/GDM/ 2017-18/GDBH-3	Sericite Schist	0.31
GDBH-3	146.96	147.6	CS/02/ JBP/GDM/ 2017-18/GDBH-3	Manganiferous quartzite	5.77
GDBH-3	146.91	147.427	CS/03/ JBP/GDM/ 2017-18/GDBH-3	Mn Ore	31.26
GDBH-3	147.427	147.69	CS/04/ JBP/GDM/ 2017-18/GDBH-3	Mn Ore	28.82
GDBH-3	147.69	147.94	CS/05/ JBP/GDM/ 2017-18/GDBH-3	Phyllite	1.25
GDBH-4	128.11	128.36	CS/01/ JBP/GDM/ 2017-18/GDBH-4	Sericite Schist	0.07
GDBH-4	128.36	128.52	CS/02/ JBP/GDM/ 2017-18/GDBH-4	Manganiferous quartzite	8.08
GDBH-4	128.4	128.6	CS/03/ JBP/GDM/ 2017-18/GDBH-4	Mn Ore	30.07
GDBH-4	128.6	128.85	CS/04/ JBP/GDM/ 2017-18/GDBH-4	Phyllite	0.22
GDBH-5	122.56	122.81	CS/01/ JBP/GDM/ 2017-18/GDBH-5	Sericite Schist	0.30
GDBH-5	122.81	123.9	CS/02/ JBP/GDM/ 2017-18/GDBH-5	Manganiferous quartzite	5.14
GDBH-5	122.86	123.164	CS/03/ JBP/GDM/ 2017-18/GDBH-5	Mn Ore	20.23
GDBH-5	123.164	123.414	CS/04/ JBP/GDM/ 2017-18/GDBH-5	Mn Ore	21.88

GDBH-5	123.414	123.58	CS/05/ JBP/GDM/ 2017-18/GDBH-5	Mn Ore	34.43
GDBH-5	123.9	124.15	CS/06/ JBP/GDM/ 2017-18/GDBH-5	Phyllite	0.09
GDBH-6	141.5	141.75	CS/01/ JBP/GDM/ 2017-18/GDBH-6	Sericite Schist	0.09
GDBH-6	141.8	142.325	CS/02/ JBP/GDM/ 2017-18/GDBH-6	Manganiferous quartzite	2.32
GDBH-6	141.75	142.35	CS/03/ JBP/GDM/ 2017-18/GDBH-6	Mn Ore	28.25
GDBH-6	142.35	142.6	CS/04/ JBP/GDM/ 2017-18/GDBH-6	Phyllite	0.05
GDBH-7	139.62	139.87	CS/01/ JBP/GDM/ 2017-18/GDBH-7	Sericite Schist	0.03
GDBH-7	133.87	141.05	CS/02/ JBP/GDM/ 2017-18/GDBH-7	Manganiferous quartzite	4.85
GDBH-7	139.883	140.305	CS/03/ JBP/GDM/ 2017-18/GDBH-7	Mn Ore	40.01
GDBH-7	140.305	140.893	CS/04/ JBP/GDM/ 2017-18/GDBH-7	Mn Ore	30.06
GDBH-7	140.893	141.05	CS/05/ JBP/GDM/ 2017-18/GDBH-7	Mn Ore	36.50
GDBH-7	141.05	141.3	CS/06/ JBP/GDM/ 2017-18/GDBH-7	Phyllite	0.05
Check Samples			CHK/01/JBP/GDM/2017-18	Mn Ore	33.22
			CHK/02/JBP/GDM/2017-18	Mn Ore	41.90
			CHK/03/JBP/GDM/2017-18	Mn Ore	31.31
			CHK/04/JBP/GDM/2017-18	Manganiferous quartzite	8.82
			CHK/05/JBP/GDM/2017-18	Manganiferous quartzite	2.53
			CHK/06/JBP/GDM/2017-18	Mn Ore	22.50
			CHK/07/JBP/GDM/2017-18	Mn Ore	40.10

V.02.7 Discussion of results of chemical analysis of samples

A total of 44 nos. core samples including 7 nos Check samples, 20 nos. BRS, 10 nos. REE, 10 nos. PCS and 5 nos. of XRF samples were collected and analysed.

BRS were collected from different lithology such as granite gneiss, Schist, Phyllite and manganese ore present in and around in the study areas. Samples indicated MnO value range from 0.03% to 30.07%, high value of MnO% 11.06 to 30.07% was

found in Mn ore collected from Ukwa and Gudma pit. Values of P_2O_5 value ranging in from 0.03 to 3.83 % and Values of Fe_2O_3 vary from 0.21% to 56.21%. High values of Fe_2O_3 i.e 44.02 and 56.21 mg/kg is received in lateritic samples. Due to high amount of quartz SiO_2 value is also high and varies from 13.31% to 90.96%. High silica values i.e 90.21 and 90.96% belong to quartz vein samples. The elements like Cu, Pb, Zn, Ni, Co, Cr show sympathetic relations due to similar ionic radius. The Cu values range from 10 ppm to 740 ppm. (Annexure-I).

The sample for REE analysis (Annexure-II) collected from granite gneiss, Quartz vein, Garnetiferous quartz mica schist and soil. Though some of the REE elements viz. La and Ce are showing higher values 158.86 ppm and 306.89 ppm respectively in sample no. REE-5 (Figure V.11).

Five number of samples analysed by XRF from garnetiferous quartz muscovite schist, phyllite, pyrite bearing quartz vein, manganiferous quartzite and Mn ore. These samples was not showing any encouraging value of Mn. One sample from Mn ore showing higher value of Ba i.e 31200 mg/kg (Annexure-III).

PCS were collected from Gneiss, Schist, GQMS, Sericite schist etc. samples indicated MnO value range from 0.02% to 41.98%, high values of MnO was found in Sericite Schist samples collected from borehole GDBH-5 at hanging wall. This high value is may be due to presence of dendrite of Mn. Value of P_2O_5 ranging in from 0.04 to 2.3 % and Values of Fe_2O_3 vary from 2.97% to 6.9%. Concentration of trace element viz. Ba is ranging from 141ppm to 17341mg/kg highest value 17341mg/kg received from Manganetiferous bearing sericite chlorite schist. and value of Sr is also high in same smple i.e 724mg/kg (Annexure-IV). Core samples were collected from whole ore horizon (ore plus waste rock). Individual details of samples are mentioned in Table-VI. Here value of Mn found to varying from 0.03 to 44.94%. Lower value of Mn is received in Hanging wall and foot wall samples i.e Sericites schist and phyllite while in Manganiferous quartzite value is intermediate and in ore samples value are in higher side. Average grade of Mn in borehole GDBH-1 is 28.07%, in borehole GDBH-2 is 30.80%, in borehole GDBH-3, 19.31%, in borehole GDBH-4, 22.74%, in borehole GDBH-5, 16.39%, in borehole GDBH-6, 13.55 and in borehole GDBH-7, 20.72% was found.

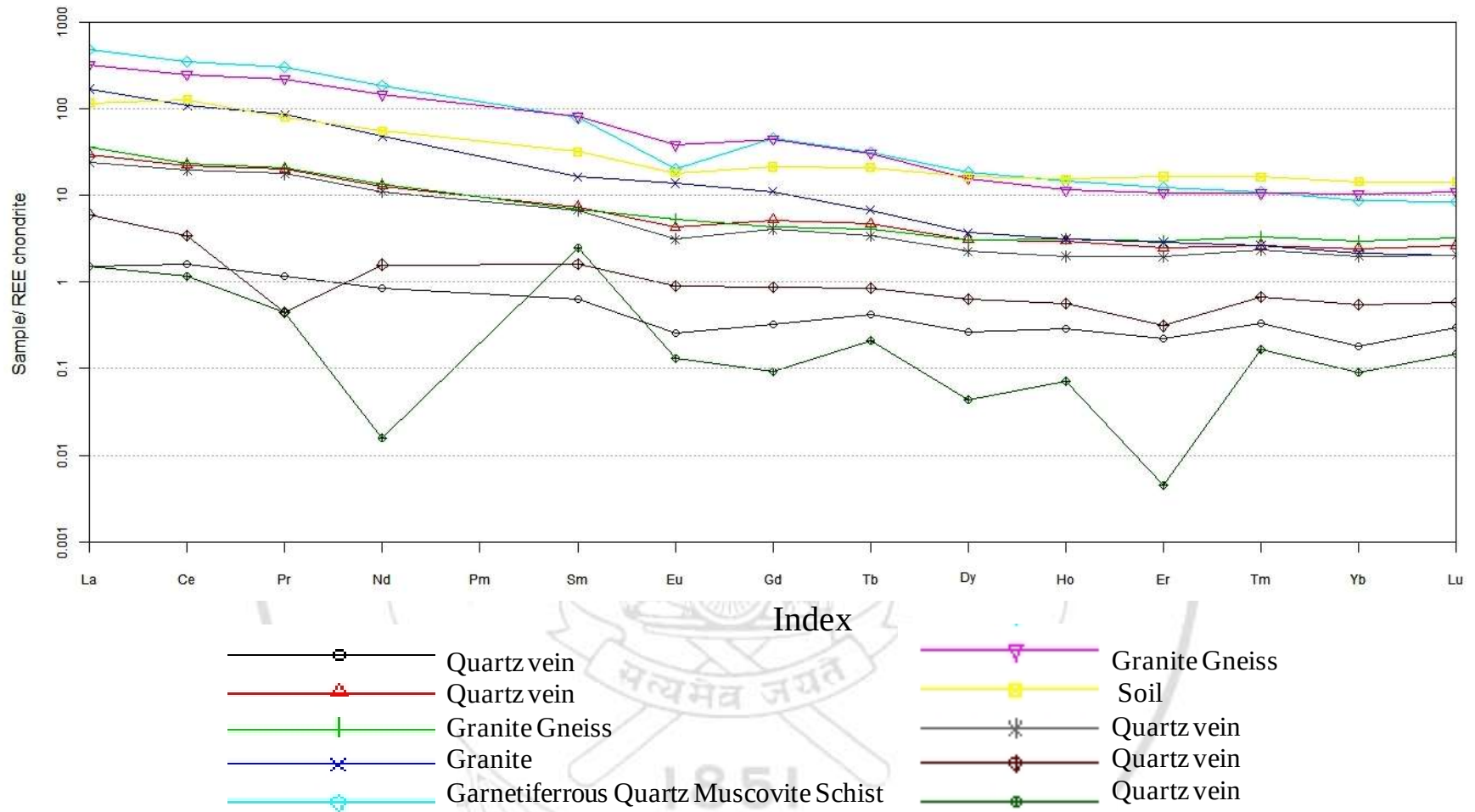


Figure V.11: REE Chondrite (Nakamura 1974) plot.

The Average values of Si for ore horizon in seven boreholes were found to be varying from 2.31% to 44.9%. Lower value received from MnQ samples while high value of Mn is obtained from Mn ore samples. Si content in ore horizon varies from 3.92 to 41.18 %. High silica is from MnQ while in ore samples silica is low compare to MnQ.

The Average values of P for ore horizon in seven boreholes were varying from 0.04% to 2.06%. Lower value of P received from MnQ samples while high value of Mn is obtained from Mn ore samples from borehole GDBH-6.

The Average values of Fe for ore horizon in seven boreholes were varying from 1.32% to 7.92%. Lower value of Fe received from MnQ samples while high value of Mn is obtained from Mn ore samples from borehole GDBH-6.

Overall grade of Mn (ore+Waste) is found 21.65% with 0.57% of P, 21.32% of Si and 4.00 % of Fe (*Annexure-V*).

V.02.8 Details of interpreted ore zone on the basis of geological investigation

During the course of G-3 stage exploration in parts of Toposheet nos. 64C/5 in Gudma block, a total of 1009.53m of drilling was done and 7 nos. of boreholes were drilled at two levels. The composite thickness of 4.38 m ore horizon was intersected in 7 different boreholes (GDBH-1, 2,3,4,5,6 & 7). The Maximum thickness of ore horizon was observed in GDBH-1 (1.38 m) and minimum thickness of ore horizon was observed in GDBH-04 (0.25 m). Details of ore zone are provided in chapter 8.

V.03 Geophysical Explorations

As per earlier geophysical exploration (Gaonkar & Das, 1995), geophysical surveys like gravity, magnetic (V.F.) and self potential methods were taken up in Ukwa (21°58'30", 80°28'W) area, Balaghat district, M.P. for finding out extension along strike for the Ukwa manganese deposit in the soil covered gap area between Ukwa deposit and Jagantola trench in the Western Block and to the east of the Eastern Block (*Figure V.12, V.13, V.14, V.15*).

The strata bound manganese bodies are located between the mica schist and the phyllites metasedimentaries. The metasedimentaries are overlying the granite gneiss basement. The bedding and the foliation trends are along ENE-WSW direction with moderate dips of about 25° to NW. The manganese horizon is exposed over a strike length of about 5 km and the thickness of the horizon ranges from 0.27 m to 6.1 m.

Two major magnetic anomalies, in the form of linear bands with strike length of about 3.5 km each and magnitude in the range of 100 to 400 gammas were picked up in the western block of this area (*Figure V.12*). The width is about 25 and 100 m for the southern and the northern anomalies respectively. The southern magnetic high was recorded over the main Ukwa mine trench, where the ore body with average width of 3 m is exposed at the surface. The anomaly shows a dominant positive high with absence of negative low to the north. This shape is attributed to the northward dip of the ore body. The gravity surveys (*Figure V.11 & V.12*) closely corroborate the results of magnetic surveys in this area. Two linear gravity anomalies corroborate with the observed linear magnetic anomalies. The geophysical surveys indicate strike continuation of the exposed manganese mineralisation under soil cover, from Ukwa trench in the east to Jagantola trench in the west, for over 1.6 km. Presence of another gravity-magnetic linear anomaly in the soil covered area to the north of the ore body exposed in the main trench indicates interesting possibilities for the presence of manganese mineralisation. S.P. surveys were also carried out in this area (*Figure V.11*). The surveys have brought out feeble but definite low values of the order of 20 to 80 mv and corroborate with the gravity magnetic response over the manganese ore body exposed in trench and also support its extension to the west over a strike length of 1.6 km. No S.P. response was recorded over the northern gravity magnetic anomaly band. This could be due to the thick soil cover and the causative body being at large depth.

A magnetic peak with narrow width was picked up over manganese pit on Tr. TU4 in the eastern block. However this anomaly was not found to continue over the next traverse located at an interval of 300 m to east. From the appearance of rapid fluctuations in magnetic values on traverses laid further to the east in this block, it is inferred that Sausar formations have gone under Deccan Trap rubble.

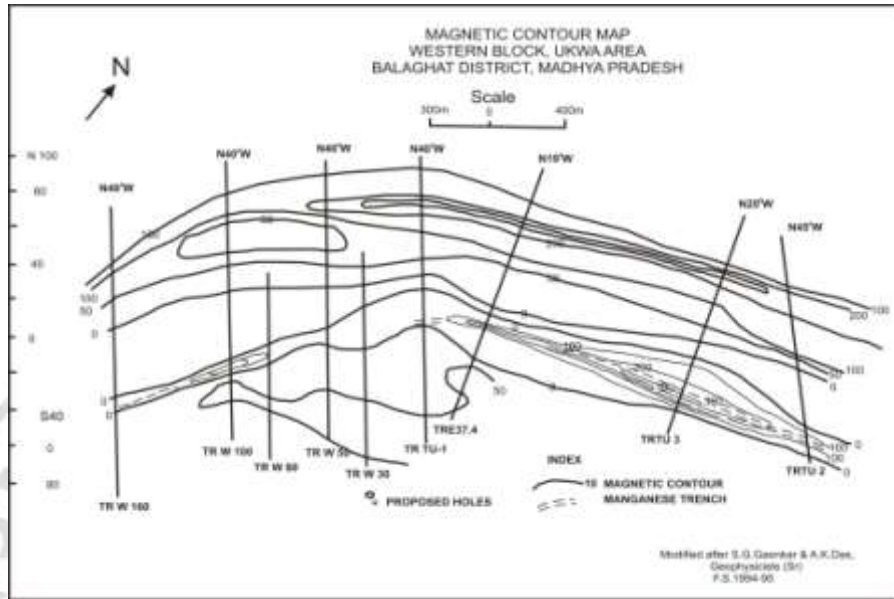


Figure V.12 : Magnetic contour map of Western Ukwa Block (Reproduced from Gaonkar & Das, 1995).

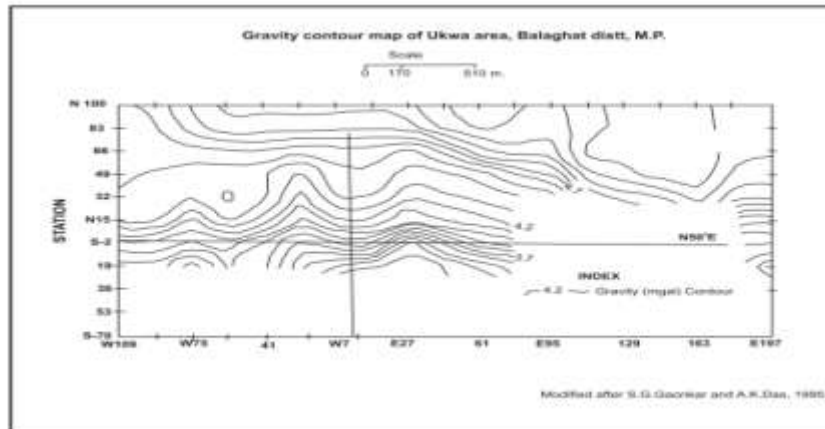
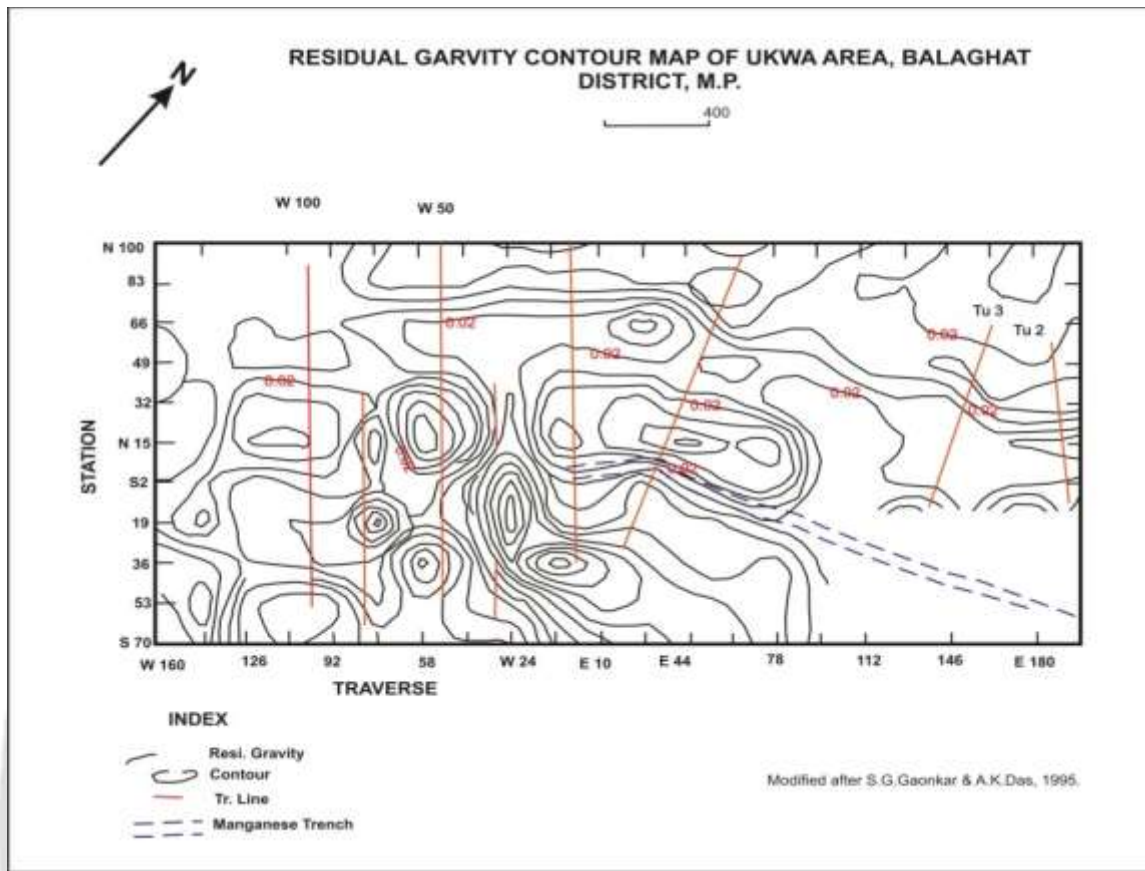
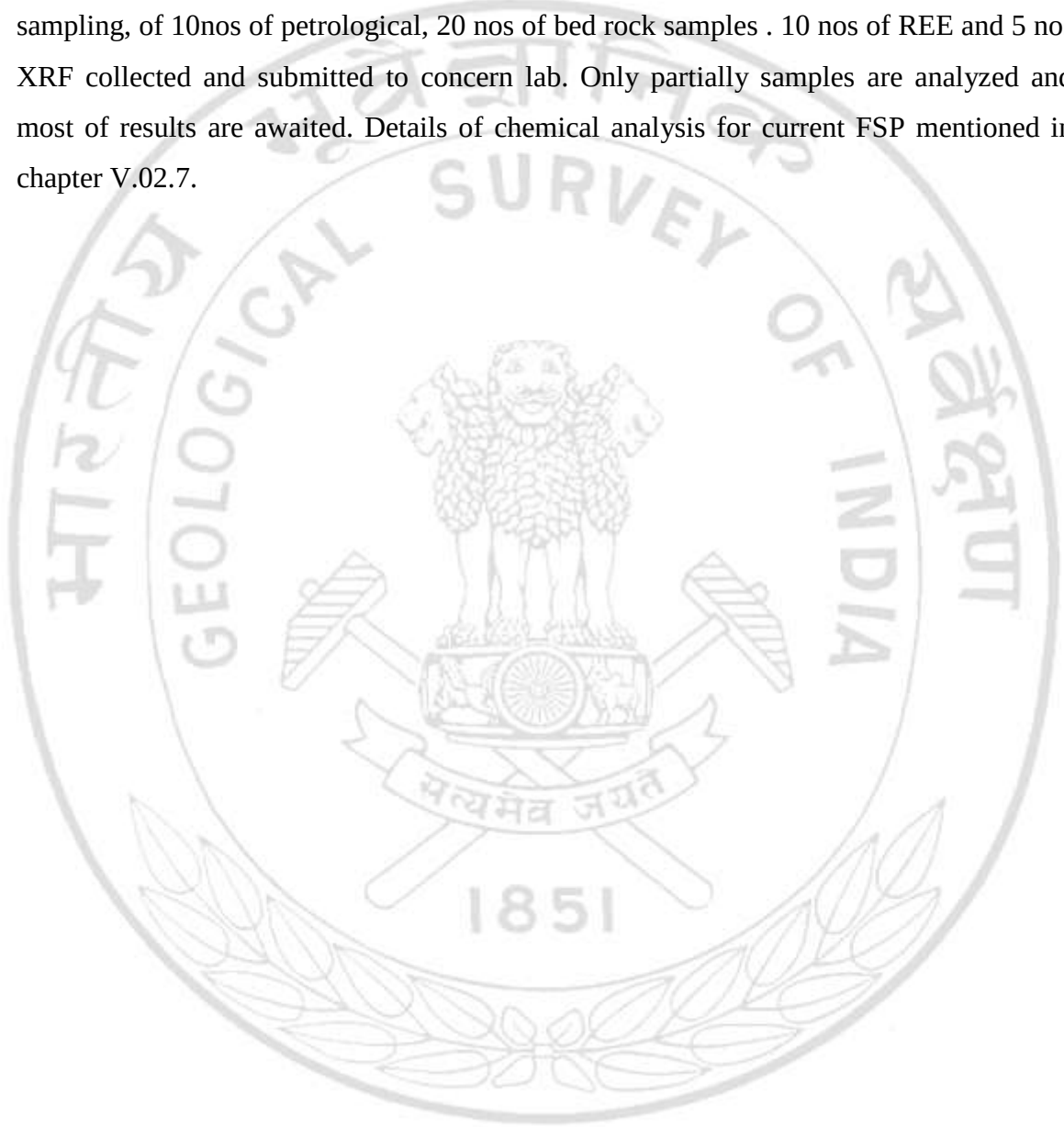


Figure V.13 : Gravity contour map of Western Ukwa Block (Reproduced from Gaonkar & Das, 1995).



V.04 Geochemical Explorations

Geochemical exploration (NGCM) have not been carried out in TS no 64C/05 so far due to inaccessibility in area (hilly terrain, dense jungle) and Naxals activities reported around. During Current FSP 44 nos of core samples, 10 nos of petrochemical sampling, of 10nos of petrological, 20 nos of bed rock samples . 10 nos of REE and 5 nos XRF collected and submitted to concern lab. Only partially samples are analyzed and most of results are awaited. Details of chemical analysis for current FSP mentioned in chapter V.02.7.



CHAPTER VI

INTEGRATION OF GEOLOGICAL, GEOCHEMICAL AND GEOPHYSICAL EXPLORATION DATA AND INTERPRETATION THERE OFF

VI.01 Creation of Geophysical, Geochemical, Geological (Lithological & Structural) and Outcrop maps on true scales

During FSP 2017-18 detailed mapping on 1:2000 scale has been carried out to plan exploration work. Earlier workers carried out geophysical work in Ukwā area which was explained in chapter V (*Figure V.12,13, 14 & 15*, references therein).

VI.02 Overlay studies in GIS (Geological, Geochemical and Geophysical maps)

No overlay studies were carried out. During current study detailed geological map on 1:2000 scale were used and digitized along with borehole locations of present work in for better representation.

VI.03 Semitransparent overlay maps

Semitransparent overlay map could not be prepared because of unavailability of map on true to the scale.

CHAPTER VII

ABIOTIC PARAMETERS

VII.01 Soil, Surface water (Pre-monsoon and post monsoon), ground water and air sampling

The present area comprises of Mansar Group of rocks which are result of low grade metamorphism of pelitic rocks in the area. The lithounit consist of phyllite, quartz-muscovite-sericite-schist, manganese ore horizon and basement biotite gneiss mainly. Residual soils are common in the area and soil profile is well developed. Gully and sheet erosion is the major kind of soil erosion in this area. On the basis of texture, soil in the area are widely classified into four classes-

A). Coarse soil: It is mainly stony and gravelly soils characterized by low-grade soils with admixture of cobbles, pebbles and gravels generally found at the foot hills and along nalas.

B). Medium soil: It is sandy in nature and generally found near the river and stream beds with 60-70 percent sand and poor plant nutrients.

C). Fine soil: It loamy soils composed mostly of sand and silt with smaller amount of clay (40%-20% concentration respectively). This is the most common type of soil in the area.

D). Very fine soil: It is mainly clayey soils characterized by sticky when wet and very hard when dry.

No surface and ground water sampling has been carried out in current FSP. Kumar et al (2016-17) carried out water analysis and reported that surface water indicates manganese rich concentration and shows low conductivity and TDS. In general the water condition in the area is good, less contaminated and drinkable. Chemical analysis of water sample indicates high value of Mn and Fe which is related to contamination of ground water by leaching from mine dump and geological formations. Rest of the values of elements is within crustal abundance limit.

Ground water condition in the area is quite good. The use of ground water is almost negligible in the area. Mica schist being the major litho unit in the country acts as good aquifer. Luxuriant growths of natural vegetation like palm and jamun trees indicate the shallow water table condition (After Kumar et al 2016-17).

No air sampling was carried out during the course of drilling in Gudma block.



CHAPTER VIII

MINERAL DEPOSIT

VIII.01 Surface indication of mineralization

A NE-SW trending ridge exposes lithounits of Mansar Group of rock viz. phyllite, quartz-muscovite-schist and Mn ore horizon (at places) in western Ukwā block. Earlier workers collected bed rock samples from manganiferous zone. The encouraging value of Mn leads to drilling in Ukwā area. At present there are no surficial outcrops present in the area. All the outcrops have been mined out in open cast mines by MOIL. In present scenario, drilling is performed here on the basis of previous work along strike extension of western Ukwā block. To have idea about nature of mineralization underground and open cast mines were visited during the regional traverses. No surficial indication such as oxidised zone was encountered.

VIII.02 Mode of occurrence

Manganese ore horizon is part of Mansar horizon. However there is transgressive relationship between lower phyllitic (calcareous) and upper ore bearing Mansar formation as calcite is found in lower part of ore horizon in the form of thin veins. Mn ore occurs as <1 cm to 15 cm thick bands interbanded with cherty to jaspery manganiferous and ferruginous quartzite and impure calcareous rock as quartz-sericite/muscovite-calcite schist (After Khan et al 1993-96). The ore is either massive in nature or is banded with mm thin quartzite and carbonate partings. Very often biotite occurs as thin layers along bedding plane. The ore is dark grey coloured and fine grained and the ore horizon at places shows fractured to brecciated nature. The interstices in brecciated and fractured zones are occupied by quartz and carbonate veinlets and encrustations. The fracturing and brecciation deteriorates the quality of the ore (After Khan et al 1993-96).

VIII.03 Nature of Mineralisation

The ore horizon is shallow, thin and pockety at the eastern end of Ukwā deposit, whereas it is thin but continuous at the western end of the deposit. During the course of drilling in strike extension of western block, ore horizon is found to be continuing along strike as well along dip, however thickness variation is observed (Table. VI). The surface and subsurface observations have clearly indicated the syn-sedimentary stratiform origin

of the ore, as evidenced by lithological association and conformable nature of ore bands with them. It shows pinch and swell structures, which may be due to combined effect of depositional feature with different phases of deformation. The culmination and depression in the levels of intersection of ore horizon along the strike and dip direction is perhaps because of combined effects of multiple deformation (Figure V.6, A, B).

VIII.04 Details of Mineralised Zone

As per earlier works Ukwá deposit had a strike length of 5.4 km in total. However, during present work, the ore horizon is found to be persisting along strike extension of western Ukwá block. Width of mineralised zone found in this block varies from 0.25 m to 1.38 m (thickness of ore horizon as found during drilling). No geophysical or geochemical exploration is carried out during current work. Details of mineralized zone are given in table no. VI below as intersected during drilling.

Table- VI Depth of intersection of mineralization in Gudma Block

Section Line	BH	Total Drill Depth in meter	Mineralized Zone (MZ) in meter		Thickness of MZ in meter
			From	To	
AB	GDBH-01	106.50	99.82	101.20	1.38
	GDBH-03	167.20	146.91	147.6	0.69
CD	GDBH-02	119.4	97.45	98	0.55
	GDBH-04	156.60	128.36	128.6	0.25
EF	GDBH-05	130	122.81	123.9	1.09
	GDBH-06	160	141.75	142.35	0.60
	GDBH-07	170	139.87	141.05	1.18

VIII.05 Alteration zones and its relevance with mineralization

As there is no role of metamorphic fluids in deposition of ore so no alteration zones are expected in the area. Post depositional deformations have resulted minor changes in disposition of ore horizon.

VIII.06 Genesis of Mineralisation/ Genetic model for mineralisation

The composition of footwall and hanging wall rocks and association of manganeseiferous cherry quartzite, carbonate rock with phyllitic intercalation in ore horizon suggests that the total association is characteristic of chemopelitic environment of deposition. The contacts of ore horizon with footwall and hanging wall rocks are quite

sharp. Even within the ore horizon different litho units have sharp contact with each other. The sequence could be represented as below:

Hanging Wall: Quartz sericite schist

Ore horizon

Foot Wall: Sericitic phyllite and schist with calcareous intercalations

The ore horizon has conformable relationship with hanging wall and footwall rocks. It has quite distinct lithological characters mostly representing chemogenic association within chemopelitic to pelitic sequence forming footwall and hanging wall rocks (Khan and Fulmari, 1996). No genetic model is being suggested by present authors.

The Proterozoic (c. 2.0Ga) Sausar Group, India, including the manganese deposits, have been complexly deformed and metamorphosed to grades ranging from lower greenschist facies to upper amphibolite facies. This sequence consists of metamorphosed equivalents of a limestone-shale-orthoquartzite assemblage (Roy 1966, 1981) indicating a shelf environment. The presence of dolomite at the top (Bichua Fm) indicates further shallowing of the basin. Interbanded Mn oxide orebodies (braunite, bixbyite, hollandite, jacobsonite, hausmannite) and Mn silicate-oxide rocks are interstratified with metapelites and orthoquartzites (Mansar Fm) and less commonly occur as conformable lenses in carbonate rocks of the older Lohangi Fm. The protoliths of these Mn rich rocks were Mn oxide/hydroxide sediments admixed with variable amounts of Fe, Si, and Al. The Mn silicate-carbonate rocks, derived from Mn carbonate progenitor with admixed impurities, also occur as isolated pockets in the Mn oxide ore horizon of the Mansar Fm. The Mn oxide deposits are inferred to have formed on the continental shelf during a sea-level highstand, when detrital supply was minimal. Deposition above the redox interface produced Mn oxides even on the carbonate substrate (Lohangi Fm). Mn carbonate sediments in the Mansar Fm were diagenetically derived from Mn oxides by reaction with calcareous partings in isolated pools where an evaporative condition developed. (Genetic diversity of manganese deposition in the terrestrial geological record - After S. Roy et al).

CHAPTER IX

EXPLORATION BY DRILLING

IX.01 Stage of exploration by drilling

As per UNFC classification, this exploration can be classified as G-3 stage as drilling is being carried out for assessment of manganese ore potential in strike extension of western Ukwa block and to know strike extension of Manganese mineralisation in depth.

IX.01.1 Field guidelines for adoption of United Nations Framework Classification

Manganese horizons in sedimentary and metasedimentary sequences of stratiform and stratabound nature are classified for UNFC norms on the following basis:

1. Geological survey:

- (i) Mapping on 1:25,000 to 1:5,000 or larger scale with triangulation points, benchmarks, if any shown.
- (ii) Linking of maps so prepared with topogrid.
- (iii) Assessment of lithology, structure, surface mineralisation, analysis of old history of mining.

2. Geochemical survey:

- (i) Grid geochemical sampling- rock type wise and if necessary, soil domain- wise (for all metallic mineral exploration).

3. Geophysical survey:

- (i) Detailed ground geophysical work.

4. Technological:

- (i) Pitting/trenching/ drilling depending on variability;
- (ii) Selection of drilling sites best suited to unravel the lithological/ structural complexities.

5. Petrographic:

- (i) Petrographic study of rocks of the deposit and its surroundings, alterations (if any), connected with mineralisation;
- (ii) Determination of phase in which mineral of interest occur;

(iii) Mineralogical studies including paragenesis, identification of zones of oxidation and primary zones, grain size distribution, overall characteristics of useful minerals.

IX.02 Methodology of drilling with details of type of drilling

In Gudma block conventional diamond core drilling was done. In conventional type of drilling after each cutting run core barrel assembly was lift to recover core samples. The drill bit used here is composed of group of small, industrial grade diamonds set into a metallic, soft matrix. As the ground is drilled, the matrix will wear away and expose more diamonds. This bit is attached to a drill rod or core barrel assembly, which is around 10 feet in length, and then more members of pipe can be attached to the top of this so a greater depth can be drilled. Drilling work in Gudma block was done by engineering division Nagpur with conventional method and rock-drill 414 & 416 drilling unit was used.

IX.03 Borehole planning

As per the objective assigned in FSP 12nos of bore hole was proposed in Gudma block in strike continuation of western block of the U kwa deposit for a total of 1000m drilling around 130m borehole interval. Borehole had to be planned to intersect ore horizon by putting 12 bore holes that is 6 bore holes had to planned to intersect the ore body at first level (around 50m vertical depth) and 6 boreholes to intersect the ore body at second level (around 100m vertical depth). During the borehole planning it was observed that the first series of boreholes were falling under the MOIL leasehold area or where MOIL has applied for lease so it was decided to intersect ore body from out of moil lease hold area at 100m vertical depth. First two boreholes GDBH-1 & 2 were planned to intersect ore body at second level i.e 100m. Borehole GDBH-3 & 4 was planned to intersect ore body at third level i.e. 150m. After drilling it was found that ore body is pinching in dip direction. During MTR 2017-18 it was decided than third level drilling should not be done as ore body pinching in dip direction. So further three boreholes GDBH-5 to 7 was planned at second level.

IX.04 Borehole Logging

Borehole logging was carried out and results are described as summarized litholog in tabular form. GDBH-1 GDBH -2, GDBH-5, GDBH-6 and GDBH -7 are

second level boreholes while GDBH-3, and GDBH-4 are third level boreholes. All the GDBH series boreholes are 65° South easterly inclined.

BOREHOLE NO. GDBH-1:

Location	Gudma Block,	Lat/Long	21°58'2.186'' : 80°26'42.351''
Section Line No	A-B	Date of commencement	05/11/2017.
R.L at Collar	597 m	Date of completion	29/11/2017.
Total Depth	106.50 m	Vertical Depth of intersection:	90.46 m

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0	3	Yellow silty soil with rock fragments.	
3	25.80	Lateritic soil with rock fragments.	
25.80	44.10	Weathered garnetiferous quartz muscovite schist.	
44.10	74.10	Garnetiferous quartz muscovite schist with vein quartz at places.	
74.10	93.45	Chlorite quartz muscovite schist and vein quartz with rare garnets. Here concentration of chlorite is increasing with depth and concentration of garnet is decreasing with depth.	
93.45	99.82	Sericite Chlorite Schist	
99.82	101.20	Ore Horizon (Mn ore is interbanded with Mn-quartzite).	Core recovery in ore zone = 96.00%.
101.20	104.95	Phyllite	Dendrites of Mn are observed along foliation plane.
104.95	106.90	Biotite Gneiss with veins of calcite and quartz.	Very fine grained due to tectonic contact. Specks of magnetite and pyrite are observed within gneiss in disseminated form.

BOREHOLE NO. GDBH-2 :

Location Gudma Block, Lat/Long 21°57'59.4'' : 80°26'39.216''
 Section Line C-D Date of commencement 21/11/2017.
 No
 R.L at Collar 595 m. Date of completion 22/12/2017
 Total Depth 119.4 m. Vertical Depth of 88.31 m
 intersection:

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0	2.6	Yellow clayey soil.	
2.6	19.70	Lateritic soil with rock fragments.	
19.70	26.00	Weathered garnetiferous quartz muscovite schist.	
26.00	67.80	Garnetiferous quartz muscovite schist with vein quartz at places.	
67.80	91.25	Chlorite quartz muscovite schist and vein quartz with rare garnets. Here concentration of chlorite is increasing with depth and concentration of garnet is decreasing with depth.	
91.25	97.45	Sericite Chlorite Schist	
97.45	98.00	Ore Horizon (Mn ore is interbanded with Mn-quartzite).	Core recovery in ore zone = 100%.
98.00	107.20	Phyllite	Dendrites of Mn are observed along foliation plane.
107.20	119.40	Biotite Gneiss with veins of calcite and quartz.	Initially very fine grained due to tectonic contact. Specks of magnetite and pyrite are observed within gneiss in disseminated form.

BOREHOLE NO. GDBH-3:

Location Gudma Block Lat/Long 21°58'4.975":80°26'39.199"
 Section Line No A-B Date of commencement 04/12/2017
 R.L at Collar 597 m. Date of completion 30/12/2017.
 Total Depth 167.20 m. Vertical Depth of intersection: 133.14 m

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0	3.64	Grey/Khakhi soil	
3.64	18.12	Red color lateritic Soil	
18.12	38.34	Silty sandy soil with rock fragment and at places Garnetiferous quartz mica schist	
38.34	52	Sand/Silt in Garnetiferous quartz mica schist a places schist is chloritic	.
52	121.7	Garnetiferous quartz muscovite schist with vein quartz at places.	
121.7	143.5	Chlorite quartz muscovite schist and vein quartz with rare garnets.	
143.5	146.91	Sericite Chlorite Schist	
146.91	147.6	Ore Horizon (Mn ore is interbanded with Mn-quartzite).	Core recovery in ore zone = 100%.
147.6	153.71	Phyllite	Dendrites of Mn are observed along foliation plane.
153.71	164.01	Biotite Gneiss with veins of calcite and quartz.	Very fine grained due to tectonic contact.
164.01	167	Coursed grained gneiss	

BOREHOLE NO. GDBH-4:

Location Gudma Block, Lat/Long 21°58'02.4":80°26'36.8"
 Section Line No C-D Date of commencement 24/12/2017
 R.L at Collar 596 m. Date of completion 16/01/2018
 Total Depth 156.60 m. Vertical Depth of 116.33 m.
 intersection:

From	To		
0	4	Yellow clayey soil.	
4	13	Lateritic soil with rock fragments.	
13	14	Weathered garnetiferous quartz muscovite schist.	
14	18.5	Garnetiferous quartz muscovite schist with vein quartz at places.	
18.5	23	Chlorite quartz muscovite schist and vein quartz with rare garnets. Here concentration of chlorite is increasing with depth and concentration of garnet is decreasing with depth.	
23	25	Sericite Chlorite Schist	
25	30	Ore Horizon (Mn ore is interbanded with Mn-quartzite).	
30	41.6	Phyllite	
41.6	99.05	Biotite Gneiss with veins of calcite and quartz.	
99.05	107.25	Quartz muscovite schist with rare garnets.	
107.25	122.7	Quartz muscovite schist	
122.7	128.36	Sericite schist(With Chlorite increasing downward)	
128.6	132.02	Sericite schist (with Mn ore). Ore Horizon (Mn ore is interbanded with Mn-quartzite).	Core recovery in ore zone = 100%..
132.02	147	Phyllite	Dendrites of Mn are observed along foliation plane.
147	156.6	Biotite Gneiss with veins of calcite and quartz.	Very fine grained due to tectonic contact. Specks of magnetite and pyrite are observed within gneiss in disseminated form.

BOREHOLE NO. GDBH-5:

Location Gudma Block, Lat/Long 21°57'57.5":80°26'35.9"
 Section Line No E-F. Date of commencement 08/01/2018.
 R.L at Collar 594 m. Date of completion 21/01/2018
 Total Depth 130 m Vertical Depth of 111.30 m
 intersection:

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0	12.10	Yellow clayey soil.	
12.10	18.05	Lateritic soil with rock fragments.	
18.05	21.20	Weathered garnetiferous quartz muscovite schist.	
21.20	69.80	Garnetiferous quartz muscovite schist with vein quartz at places.	
69.80	116.30	Chlorite quartz muscovite schist and vein quartz with rare garnets. Here concentration of chlorite is increasing with depth and concentration of garnet is decreasing with depth.	
116.30	122.81	Sericite Chlorite Schist	
122.81	123.90	Ore Horizon (Mn ore is interbanded with Mn-quartzite).	Core recovery in ore zone = 100%.
123.90	129.78	Phyllite	Dendrites of Mn are observed along foliation plane.
129.78	130.40	Biotite Gneiss with veins of calcite and quartz.	Very fine grained due to tectonic contact. Specks of magnetite and pyrite are observed within gneiss in disseminated form.

BOREHOLE NO. GDBH-6:

Location Gudma Block Lat/Long 21°57'54.405":
80°26'31.671"

Section Line G-H Date of commencement 22/01/2018
No

R.L at Collar 592.6 m Date of completion 19/02/2018

Total Depth 160 m Vertical Depth of 128.46 m
intersection:

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0	7	Red/Khakhi soil	
7	15	Red color silty soil	
15	25.5	Red color Soil with sand and rock fragments	
25.5	29	Sand with garnetiferous	
29	36.5	Garnetiferous Quartz Muscovite Schist with mangnatite	
36.5	69.9	Greenish color coarsed grained Garnetiferous Quartz Muscovite Schist	
69.9	104.5	Greenish color medium grained Garnetiferous Quartz Muscovite Schist	
104.5	135.1	Greenish color medium grained Garnetiferous Quartz Muscovite Schist	
135.1	141.75	Sericite schist(With Chlorite increasing downward)	
141.75	142.35	Sericite schist (with Mn ore). Mn ore is brecciated and interbanded with quartzite.	Core recovery in Ore zone = 100%.
142.35	150.54	phyllite	Dendrites of Mn are observed along foliation plane.
150.54	160.05	Fine grained gneissic (Biotite)	Very fine grained due to tectonic contact. Specks of magnetite and pyrite are observed within gneiss in disseminated form.

BOREHOLE NO. GDBH-7:

Location Gudma Block, Lat/Long 21°57'49.4'':80°26'27.2''
 Section Line No I-J Date of commencement 23/02/2018.

R.L at Collar 589.7 m Date of completion 13/03/2018

Total Depth 170 m. Vertical Depth of 126.76 m
 intersection:

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0	15	Red/Khakhi soil	
15	35.5	Soil and Garnetiferous Quartz Muscovite Schist	
35.5	65.8	Garnetiferous Quartz Muscovite Schist	
65.8	129.8	Garnetiferous Quartz Muscovite Schist at places it is chloritic	.
129.8	133	Garnetiferous Quartz Muscovite Schist with mangnatite	
133	139.87	Sericite schist	
139.87	141.05	Mn ore interbanded with quartzite.	Core recovery = 100%.
141.05	146.9	phyllite	Dendrites of Mn are observed along foliation plane.
146.9	170	Fine grained gneissic material	Very fine grained due to tectonic contact. Specks of magnetite and pyrite are observed within gneiss in disseminated form.

IX.05 Core Recovery Percentage

Core recovery is the borehole core recovery percentage of solid, cylindrical, pieces of rock core. In Gudma block core recovery percentage varies from top to bottom. From top to 20m depth average core recovery is about 0-5%, from 20 to 40m it is about 50% and after 40m to 128m in Schistose rock it is about 90 to 100%. At the mineralised zone recovery varies from 90 to 100%. At footwall side core recovery is 40 to 50% and after phyllite in gneissic rock it is about 95 to 100%.

IX.06 Geophysical logging of borehole and correlation with borehole Geology

Geophysical borehole loggings have been carried out in boreholes GDBH-1 and GDBH-3, Gudma block, Balaghat, Madhya Pradesh by deploying Microllogger unit and acquired the data of Self Potential, Single Point Resistance, Natural Gamma and Resistivity as per requisition received from Geologist of the project for F.S. 2017-18. Resistance / Resistivity logs in conjunction with SP log prove useful in detection of mineralized zones.

1) BOREHOLE No. - GDBH-1

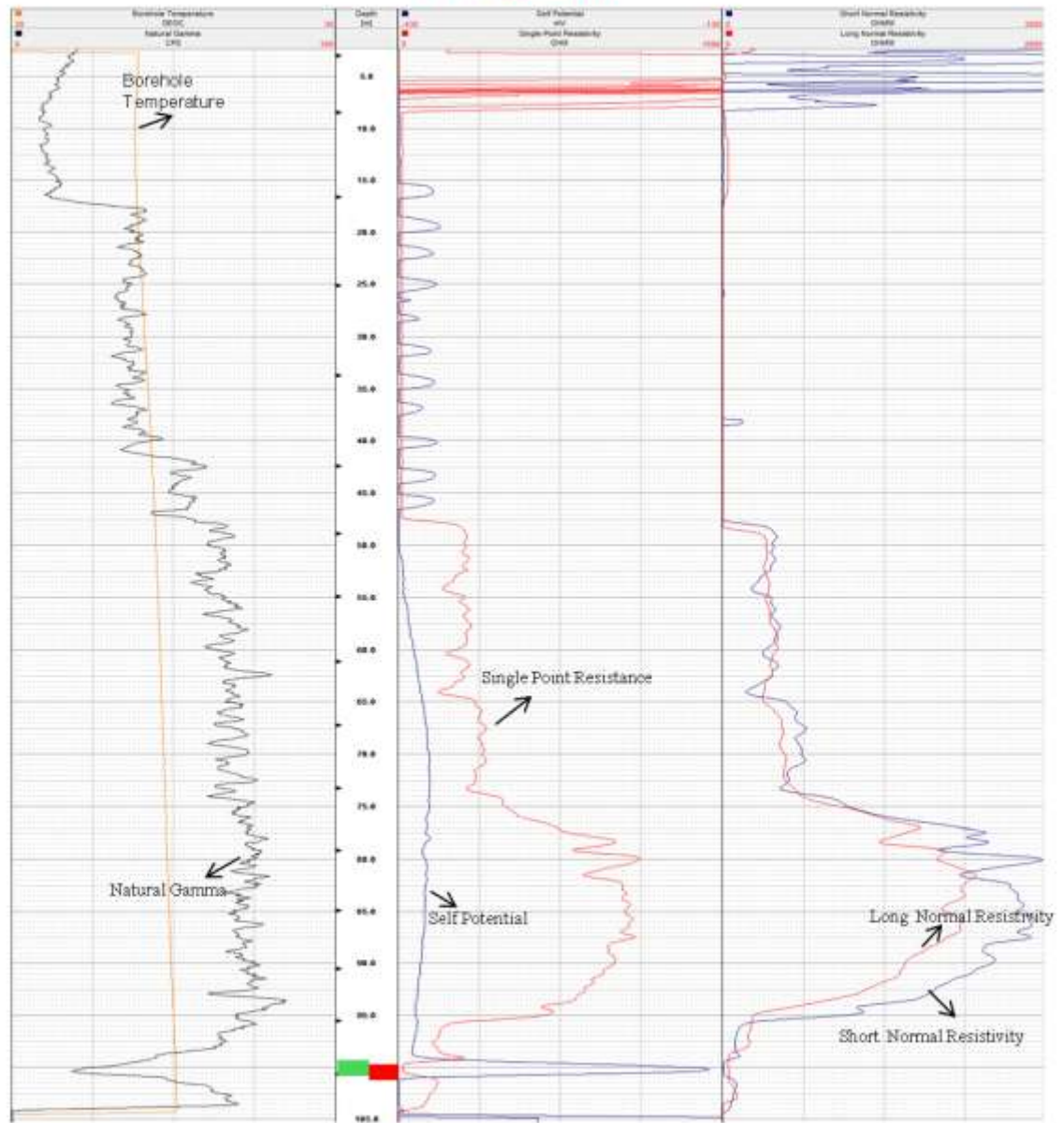
This borehole was logged on 29-11-2017 up to depth of 105.01 m and as per analysis and interpretation of the data of Self Potential, Single Point Resistance, Natural Gamma and Resistivity, one conductive zone has been identified between 99.40-100.80 m.

2) BOREHOLE No. - GDBH-3

This borehole was logged on 30-12-2017 up to depth of 166.53 m and as per analysis and interpretation of the data of Self Potential, Single Point Resistance, Natural Gamma and Resistivity, one conductive zone has been identified between 147.25-148.30 m

Figure.IX.1

**GEOPHYSICAL LOG OF BOREHOLE NO. GDBH-1, GUDMA BLOCK,
BALAGHAT, MADHYA PRADESH
(F.S. 2017 – 18)**

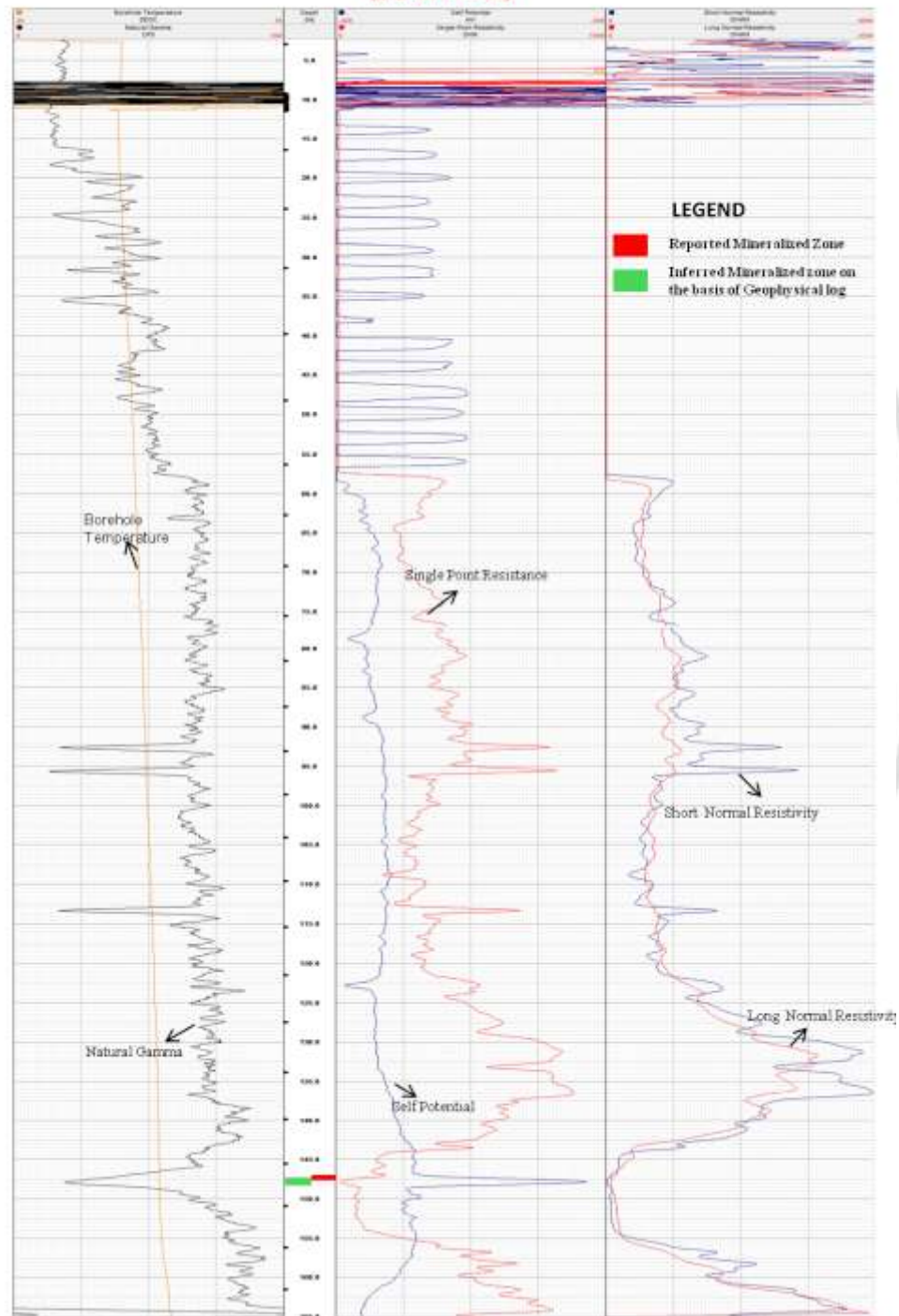


LEGEND

- Reported Mineralized Zone
- Inferred Mineralized zone on the basis of Geophysical log

Figure.IX.2

**GEOPHYSICAL LOG OF BOREHOLE NO. GDBH-3, GUDMA BLOCK,
BALAGHAT, MADHYA PRADESH
(F.S. 2017-18)**



IX.07 Mineralogy of the ore zone

Manganese ore deposits occur mainly as metamorphosed bedded sedimentary deposits associated with Gondite Series. Mn ore occurs as <1 cm to 15 cm thick bands interbanded with cherty to jasper rich manganese and ferruginous quartzite and impure calcareous rock as quartz sericite/muscovite-calcite schist (*Figure IX.3*). Apart from manganese and ferruginous quartzite and calcareous rock, Mn ore is also interbanded occasionally with sericitic phyllite/schist, particularly at the lower portion of the horizon (*Figure IX.4*). The ore is either massive in nature or is banded with mm thin quartzite and carbonate partings. Brecciation is also very common in ore horizon. Very often biotite occurs as mm-thin layers along bedding plane. The ore is dark grey coloured and fine grained. Common ore which found in Gudma Block are Psilomelane, Braunitz, Rhodochrosite, Hollandite, Bixbyite and Hausmannite.



Figure : IX.3 Interbanding of Manganese Ore and Manganeseiferous Quartzite in GDBH-7.



Figure : IX.4 Dendrites of Manganese in Phyllite at footwall due to secondary enrichment , Borehole GDBH-6.

IX.08 Borehole deviation test and methodology

Borehole deviation test was carried out using etching method by hydrofluoric acid (HF). Test tubes were inserted at every -30 m RL for an hour with HF. Once test tubes were etched; angles were measured and deviation angle was determined (*Figure. IX.5*).

GDBH-2: Deviation of -1° to 4° was observed in this borehole.

FROM	TO	ANGLE MEASURED	ACTUAL ANGLE
0	39	71°	65°
39	78	72°	66°
78	119	74°	69°

GDBH -3: Deviation of -1.5° to 4° was observed in this borehole.

FROM	TO	ANGLE MEASURED	ACTUAL ANGLE
0	55	74°	69°
55	110	70°	63.5°
110	166	73°	67.5°

GDBH -4: Deviation of -1.5° to 2.5° was observed in this borehole.

FROM	TO	ANGLE MEASURED	ACTUAL ANGLE
0	48	70°	63.5°
48	96	69°	62.5°
96	150	73°	67.5°

GDBH -5: Deviation of -3° to 3° was observed in this borehole.

FROM	TO	ANGLE MEASURED	ACTUAL ANGLE
0	45	73°	67°
45	90	71.5°	66°
90	130	72°	65.5°

GDBH -6: Deviation of 0.5^0 to 4^0 was observed in this borehole.

FROM	TO	ANGLE MEASURED	ACTUAL ANGLE
0	53	71.5°	65.5°
53	106	72.5°	67°
106	159	74°	69°

GDBH -7: Deviation of -2.5^0 to 3^0 was observed in this borehole.

FROM	TO	ANGLE MEASURED	ACTUAL ANGLE
0	56	72°	66°
56	112	69°	62.5°
112	169	73.5°	68°

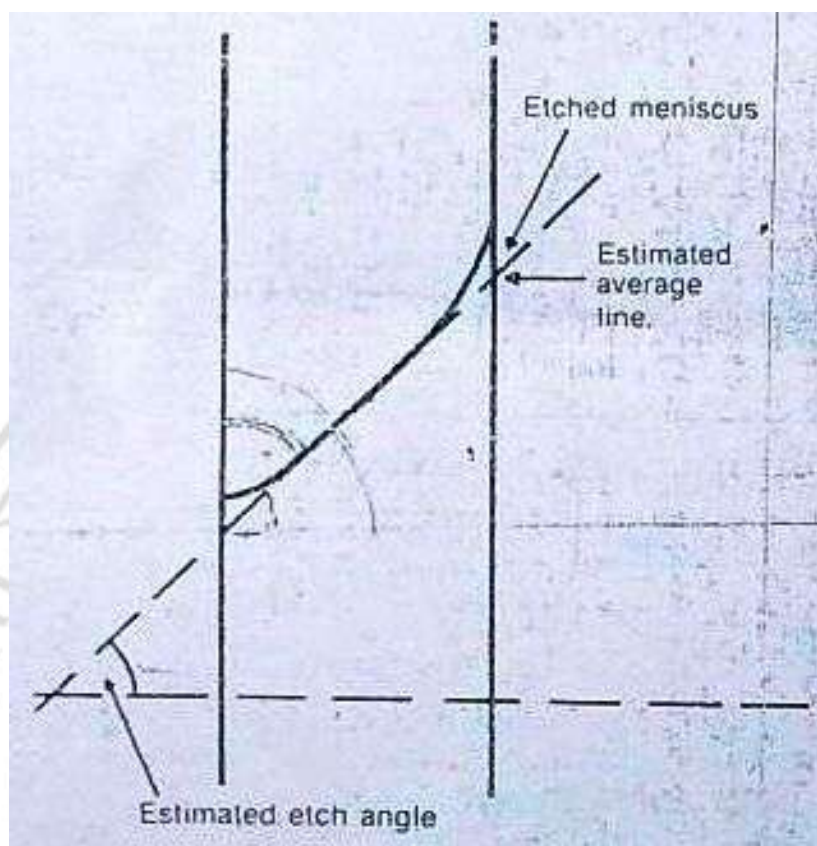


Figure IX.5 : Diagram showing etch angle measurement from test tube (D.A. Berkman,)

IX.09 Methodology of the ore zone sampling

For sample collection methodology of earlier workers (Khan et al 1993-96) has been followed. Here Mn ore and Mn quartzite has been sampled independently from ore horizon. For Mn ore sample length of 25cm has been used , though at places it is less than 25 cm due to remaining sample left after sampling. For Mn quartzite a single sample is prepared from whole ore horizon in each borehole. Samples from adjacent hanging wall and footwall have also been collected along with samples from ore horizon (Mn ore samples and Manganiferous Quartzite sample) to see the dispersion of Manganese. For hanging wall and foot wall sample length 25 cm was kept.

A total number of 44 Core samples have been prepared out of 100sample due to thinning of ore horizon. Samples were prepared as per run of drilling.

IX.10 Sample preparation, Chemical analysis and laboratory procedures

Samples from ore zone were cut into two halves. One half of sample was pulverized up to -120# size. Other one was sent to core library. Pulverized sample was thoroughly homogenized after preparation of sample in powder form. Conning and quartering had been done after homogenization and sample were collected into two equal halves. One halve of sample was sent to chemical lab, Nagpur and another one was preserved for future use. Necessary precautions were taken to avoid contamination of the samples.

IX.11 Check samples

Seven number of check samples were submitted from each boreholes for analysis.

IX.12 Details of Ore Zones of the boreholes

The details of the ore zone intersected in Gudma Block of each borehole is given below in *Table -VII*.

Table-VII The details of the ore zone intersected in Gudma Block of each borehole

Section Line	Borehole no.	Mineralized Zone (MZ) in meter		Thickness of MZ in meter	Mineralization
		From	To		
AB	GDBH-01	99.82	101.20	1.38m	Mn ore inter banded with Manganiferous Quartzite
CD	GDBH-02	97.45	98	0.55m	do
AB	GDBH-03	146.91	147.6	0.69m	do
CD	GDBH-04	128.36	128.6	0.24m	do
EF	GDBH-05	122.81	123.9	1.09m	do
GH	GDBH-06	141.75	142.35	0.60m	do
IJ	GDBH-07	139.87	141.05	1.18m	do

CHAPTER X

ORE BENEFICIATION STUDY

Ore beneficiation is a process for extractive metallurgy which improves the economic value of the ore by removing the gangue minerals, results in a higher grade product (concentrate) and a waste stream (tailings).

During current FSP no beneficiation studies has been carried out. For resource estimation bulk density of previous workers will be used.

During FSP 2016-17 about 23kg of manganese ore core sample from Ukwá western block was submitted to IBM, Nagpur for beneficiation and bulk density study. The objective of beneficiation study was to develop suitable beneficiation process route for the production of manganese concentrate for industrial use. The sample consisted major amount of braunite and quartz with subordinate amount of psilomelene, cryptomelene and carbonate (rhodochrosite, dolomite and calcite), minor amount of garnet and pyrolucite and very minor amount of feldspar (plagioclase, microcline), jacobsonite, amphibole, mica, pyroxene, hausmanite & rhodonite. Hollandite, tourmaline, clay, epidote, hematite and goethite/limonite are noticed in trace amount in the sample.

The sample assayed 22.26%Mn, 6.06%Fe(T), 41.21%SiO₂ 4.60%Al₂O₃ 3.95% CaO, 1.42%BaO, 0.98%MgO, 0.78%Na₂O, 1.40%K₂O, 0.15% TiO₂ and 0.24% P.

The bulk density of the manganese ore sample was found to be 2.09 T/m³. The beneficiation process done by gravity separation employing jigging and tabling followed by magnetic separation. The sample is amenable to beneficiation and the concentrates obtained may find use on blending.

CHAPTER XI

GEOTECHNICAL STUDIES ON BOREHOLE CORE SAMPLES OF MINERALISED ZONE, HANGING WALL AND FOOTWALL SIDE

No geotechnical studies were carried out during current FSP; however RQD analysis of core samples was done during borehole logging. Details of RQD analysis is given below:

XI.01 RQD (Rock quality designation studies): RQD analysis of cores collected were analyzed. Methodology used is explained in figure below.

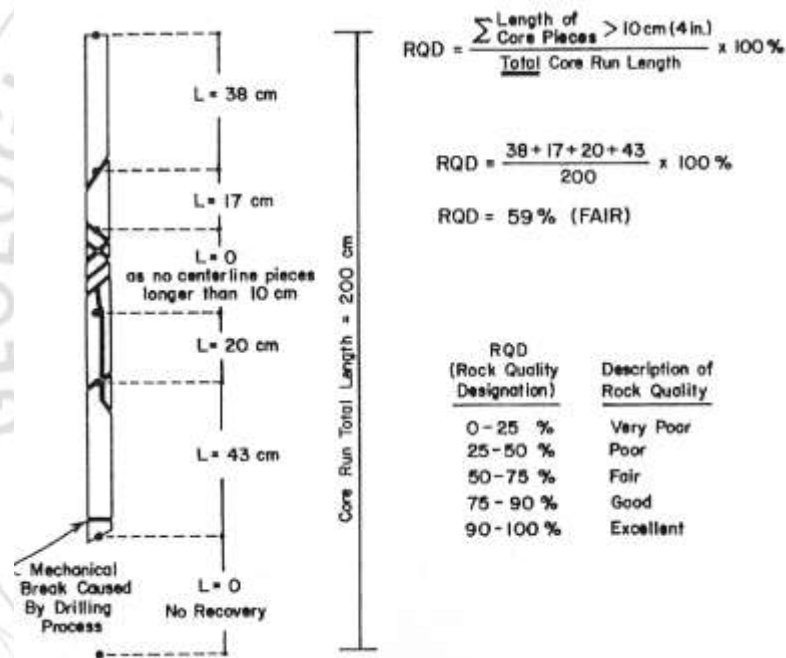


Figure XI.1: Diagram illustrating RQD calculation (D.U. Deere and D.W. Deere, 1988).

Data obtained after analysis is described in tabular form in table-IX. RQD value of only core samples after soil is mentioned below (Run with ore horizon is marked in bold).

Table- VIII Details of RQD

GDBH-1				GDBH-1		
From (m)	To (m)	RQD (%)		From (m)	To (m)	RQD (%)
25.80	44.40	0		77.25	78.05	100
44.40	45.80	56.42		78.05	80.85	29.64
45.80	47.30	92.06		80.85	84.00	56.19
47.30	48.75	66.89		84.00	87.15	88.88
48.75	51.25	42.40		87.15	90.30	91.11
51.25	53.25	98.04		90.30	93.45	75.23
53.25	55.25	65.5		93.45	96.60	51.74
55.25	57.25	79.0		96.60	99.80	64.00
57.25	60.30	91.47		99.80	100.30	26.00
60.30	60.59	34.48		100.30	100.80	0
60.59	63.15	81.64		100.80	101.20	0
63.15	66.30	59		101.20	102.70	0
66.30	68.05	74.87		102.70	104.45	0
68.05	71.20	81.58		104.45	104.95	20
71.20	74.10	86.55		104.95	106.90	95
74.10	77.25	93.33				
GDBH-2						
From (m)	To (m)	RQD (%)		From (m)	To (m)	RQD (%)
19.50	24	0		40.40	43.60	71.56
24.00	26.00	24		43.60	46.60	53.33
26.00	27.40	41.4		46.60	49.50	37.93
27.40	30.00	53.84		49.50	51.00	45.33
30.00	33.00	92		51.00	54.00	59.00
33.00	35.85	69.47		54.00	57.10	66.12
35.85	38.85	68.00		57.10	60.20	63.54
38.85	40.40	30.32		60.20	62.75	55.29

GDBH-2						
From (m)	To (m)	RQD (%)		From (m)	To (m)	RQD (%)
62.75	64.50	14.85		97.15	98.00	27.00
64.50	67.50	59.33		98.00	100.40	0
67.50	70.50	39.33		100.40	102.00	0
70.50	72.80	52.17		102.00	103.70	0
72.80	76.00	35.00		103.70	105.30	0
76.00	79.20	67.50		105.30	106.30	0
79.20	82.30	79.35		106.30	108.40	38.00
82.30	85.45	94.28		108.40	111.60	90.93
85.45	88.65	86.25		111.60	113.10	85.33
88.65	91.85	56.87		113.10	116.20	89.03
91.85	94.00	84.65		116.20	119.40	97.5
94.00	97.15	84.44				
GDBH-3						
From (m)	To (m)	RQD (%)		From (m)	To (m)	RQD (%)
48.5	50	6.67		85.5	88.8	75.15
50	52	23.00		88.8	91.5	71.85
52	55	26.33		91.5	94.3	50.71
55	58	28.67		94.3	97.4	75.81
58	59	0.00		97.4	100.5	74.52
59	62.2	55.62		100.5	102.5	64.00
62.2	65.4	52.19		102.5	105.7	50.31
65.4	68.6	57.50		105.7	108.9	48.75
68.6	71.8	63.44		108.9	112.1	26.25
71.8	75	70.94		112.1	115.3	73.44
75	78.2	47.19		115.3	118.5	70.94
78.2	81.4	67.81		118.5	121.7	65.62
81.4	83.4	66.50		121.7	124.9	62.50

83.4	85.5	61.90		124.9	128.1	84.69
128.1	130.7	77.31		148.6	149.8	0.00
130.7	133.9	66.25		149.8	151.4	6.88
133.9	136.5	58.08		151.4	153	53.13
136.5	139.7	56.25		153	154.6	93.75
139.7	142.8	85.16		154.6	157.8	76.25
142.8	146	68.44		157.8	161	68.00
146	147.6	0.00		161	164	85.63
147.6	148.6	0.00		164	167.2	26.25
GDBH-4						
From (m)	To (m)	RQD (%)		From (m)	To (m)	RQD (%)
0	41.6	0.00		101.05	104.15	28.39
41.6	45	0.00		104.15	107.25	39.68
45	46.5	33.00		107.25	110.3	37.38
46.5	49.4	66.89		110.3	112.3	11.50
49.4	52.6	48.00		112.3	114.6	6.09
52.6	55.8	29.06		114.6	116.2	17.50
55.8	58.9	38.71		116.2	119.2	29.67
58.9	62.1	56.88		119.2	119.7	0.00
62.1	65.1	65.00		119.7	122.7	61.33
65.1	68.2	25.48		125.8	128.6	28.39
68.2	71.4	30.94		128.6	130.8	61.43
71.4	74.6	68.44		130.8	132.9	5.00
74.6	77.8	39.06		132.9	135.4	5.71
77.8	81	68.44		135.4	137.4	0.00
81	84.2	67.81		137.4	140.6	23.00
84.2	87.4	68.75		140.6	143.8	92.19
87.4	90.6	60.00		143.8	147	93.13
90.6	93.8	75.00		147	150.2	100.00
93.8	95.85	49.76		150.2	153.4	93.48
95.85	99.05	56.88		153.4	156.6	79.00

99.05	101.05	37.50				
GDBH-5						
From (m)	To (m)	RQD (%)		From (m)	To (m)	RQD (%)
18.10	19.10	0		85.40	88.20	53.57
19.10	20.10	0		88.20	91.40	58.437
20.10	21.25	0		91.40	94.00	35.76
21.25	24.25	45.33		94.00	95.80	18.33
24.25	27.45	59.37		95.80	97.00	19.16
27.45	30.65	59.68		97.00	100.15	78.73
30.65	33.85	53.75		100.15	102.50	80.00
33.85	35.20	33.10		102.50	103.20	78.57
35.20	38.45	78.43		103.20	106.20	94.00
38.45	40.00	25.80		106.20	109.20	80.66
40.00	43.10	76.45		109.20	112.20	96.33
43.10	44.80	41.76		112.20	115.10	91.37
44.80	46.00	0		115.10	118.30	53.43
46.00	49.20	56.56		118.30	119.10	27.50
49.20	51.80	0		119.10	122.30	76.87
51.80	55.00	53.43		115.10	118.30	53.43
55.00	58.20	37.50		118.30	119.10	27.50
58.20	61.20	15.00		119.10	122.30	76.87
61.20	63.40	39.00		122.30	122.90	65.00
63.40	66.60	49.00		122.90	123.50	9.66
66.60	69.70	56.12		123.50	123.90	0
69.70	72.90	42.50		123.90	125.10	0
72.90	76.00	75.48		125.10	126.60	0
76.00	79.20	86.25		126.60	128.10	0
79.20	82.20	74.66		128.10	129.10	0
82.20	85.40	89.00		129.10	130.40	32.30

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GDBH-6						
From (m)	To (m)	RQD (%)		From (m)	To (m)	RQD (%)
0	27.7	0.00		101.5	104.5	50.00
27.7	29	10.77		104.5	107.6	69.67
29	32.1	17.10		107.6	110.6	54.19
32.1	35.1	41.33		110.6	113.7	37.67
35.1	36.5	15.71		113.7	116.9	55.16
36.5	39.7	53.44		116.9	120	76.25
39.7	42.9	74.69		120	123.2	67.74
42.9	45	64.76		123.2	126.3	84.69
45	48.2	28.44		126.3	129.4	85.81
48.2	51.3	84.19		129.4	132.5	83.23
51.3	54.4	40.97		132.5	135.1	92.90
54.4	57.4	55.67		135.1	138.3	79.62
57.4	60.6	62.50		138.3	141.5	89.06
60.6	63.7	51.29		141.5	141.9	65.00
63.7	66.8	78.06		141.9	142.05	0.00
66.8	69.9	55.48		142.05	142.2	0.00
69.9	73	23.55		142.2	142.35	0.00
73	76.2	77.18		142.35	144.1	0.00
76.2	79.4	64.38		144.1	145.8	24.00
79.4	82.5	72.90		145.8	147.7	0.00
82.5	85.6	86.45		147.7	150.45	0.00
85.6	88.7	83.87		150.45	153.65	43.27
88.7	91.9	59.37		153.65	156.85	92.50
91.9	95.1	70.31		156.85	160.05	87.81
95.1	98.3	66.88				

GDBH-7						
From (m)	To (m)	RQD (%)		From (m)	To (m)	RQD (%)
17.50	35.50	0		107.40	110.60	75.62
35.50	38.50	14.00		110.60	113.80	83.75
38.50	41.60	17.41		113.80	117.00	86.25
41.6	43.9	67.39		117.00	120.20	79.37
43.9	47.1	69.37		120.20	123.40	79.37
47.1	48	16.67		123.4	126.6	89.69
48	51.2	52.50		126.6	129.8	90.62
51.2	54.4	52.50		129.8	133	76.56
54.4	57.6	64.37		133	136.2	67.19
57.6	60.75	41.90		136.2	139.4	85.62
60.75	62.25	31.33		139.4	140.15	20.00
62.25	64.9	56.60		140.15	142.6	30.20
64.9	65.8	68.89		142.6	144.3	52.35
65.8	69	61.87		144.3	146.9	41.15
69	72.2	62.81		146.9	150.1	81.88
72.2	75.4	80.62		150.1	153.3	91.25
75.4	78.6	72.19		153.3	156.2	87.93
78.6	81.8	77.19		158.60	161.80	93.75
81.80	85.00	80.00		161.80	165.00	86.25
85.00	88.20	71.56		165.00	168.10	79.35
88.20	91.40	65.31		168.10	170.00	100
91.40	94.60	73.43				
94.60	97.80	73.43				
97.80	101.00	37.81				
101.00	104.20	58.75				
104.20	107.40	59.68				

CHAPTER XII

RESOURCE ESTIMATION

XII.01 Introduction

Resource estimation of ore resource is based on persistence of the ore horizon along its strike, dip, thickness and grade of the manganese ore. All factors have been determined in the exploration by drilling.

As current works is based on delineation of strike extension of western Ukwa block and to calculate resource; Gudma block is considered here for drilling with close intervals and resource estimation. LV section and cross sections for resource estimation have been prepared.

XII.02 Detailed description of ore zones with core recovery

The manganese ore zone comprises manganese ore interbanded with manganiferous quartzite and associated waste rock. Mn ore occurs as <1 cm to 15 cm thick bands interbanded with cherty to jaspery manganiferous and ferruginous quartzite and calcareous rock as quartz-sericite/muscovite-calcite schist. Apart from manganiferous and ferruginous quartzite and calcareous rock, Mn ore is also interbanded occasionally with sericitic phyllite/schist, particularly at the lower portion of the horizon. The ore is either massive in nature or is banded with mm to cm thin quartzite and carbonate partings. Depth of intersection of ore horizon is listed below in table no. X along with description of core recovery.

Table-IX Depth of intersection of mineralization in Gudma Block with recovery.

Section Line	BH	Total Drill Depth in meter	Mineralized Zone (MZ) in meter		Thickness of MZ in meter along the borehole	Core recovery
			From	To		
AB	GDBH-01	106.50	99.82	101.20	1.38	96.01%
	GDBH-03	167.20	146.91	147.6	0.69	100
CD	GDBH-02	119.4	97.45	98	0.55	100
	GDBH-04	156.60	128.36	128.6	0.25	100%
EF	GDBH-05	130	122.81	123.9	1.09	100
	GDBH-06	160	141.75	142.35	0.60	100
	GDBH-07	170	139.87	141.05	1.18	100

XII.03 Cut off grade consideration

Cut off grade of 10% will be used as per latest IBM Mineral Yearbook (2015). Earlier the lowest Mn grade to be considered was 35%.

XII.04 Minimum stopping width consideration

The thickness of ore horizon varies from 0.24 m to 1.38 m as observed during drilling. In general, the minimum stopping width is taken as 2.0 m for underground mining methods. However, for open cast mining the bench thickness is taken as stopping width, which varies from 2 m to 5 m. Here minimum stopping width is not considered as ore body is less than 2m.

XII.05 Correlation of ore lodes

The mineralised zone may be termed as the zone, which contains the ore mineral in traces or in visible quantity. However, the zones that contain the manganese ore and Manganiferous quartzite are considered as ore zones. Here ore horizon is marked on the basis of visual estimate. The correlation has been constructed by joining the ore zone intersection points of all the boreholes falling in first series (*Figure. XII.1*). Cross sections were prepared across the strike of the ore body for all the section lines (*Figure. XII.2 to 6*), along which boreholes were drilled.

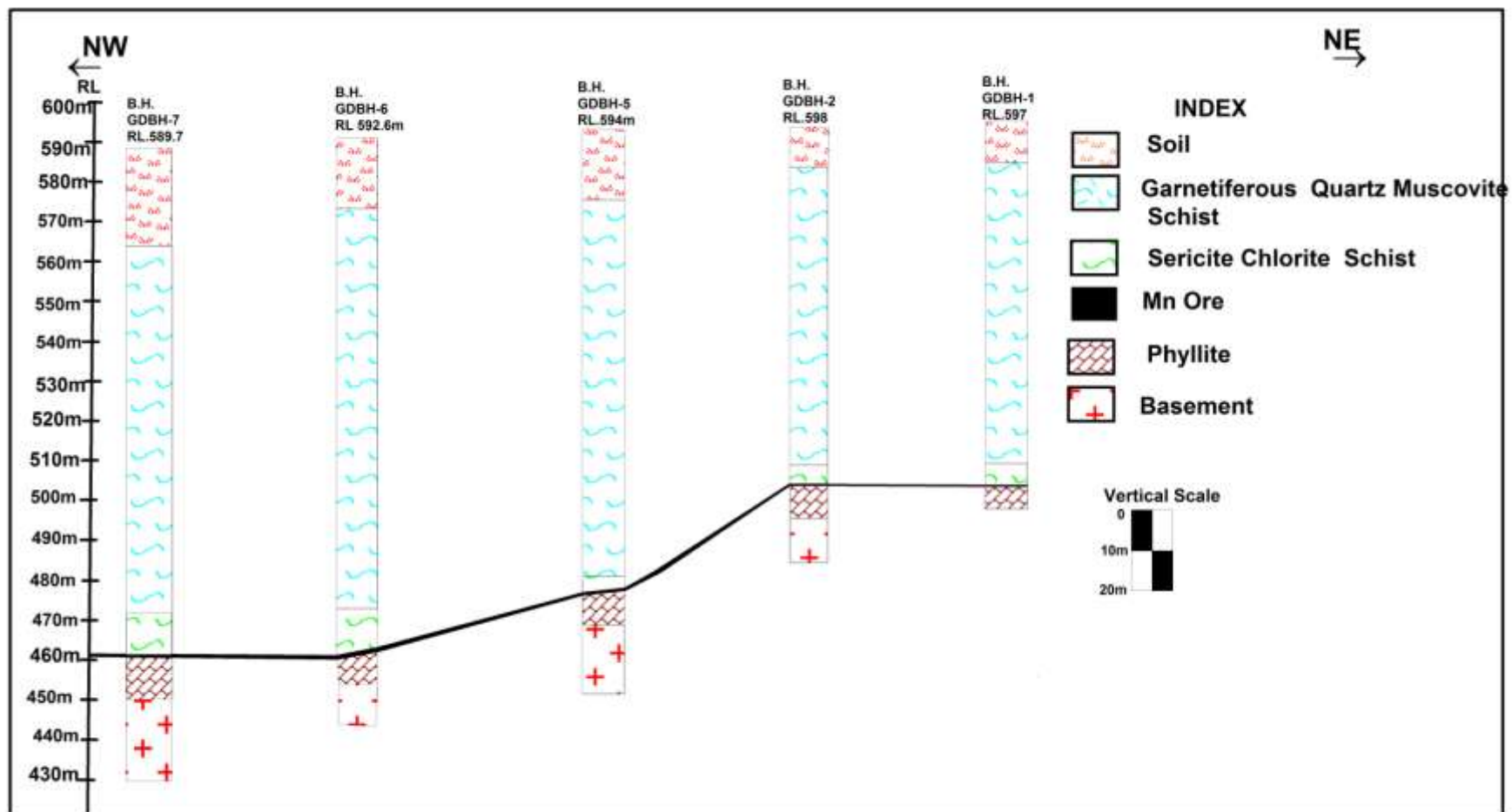


Figure XII.1 Correlation of ore load at second level.

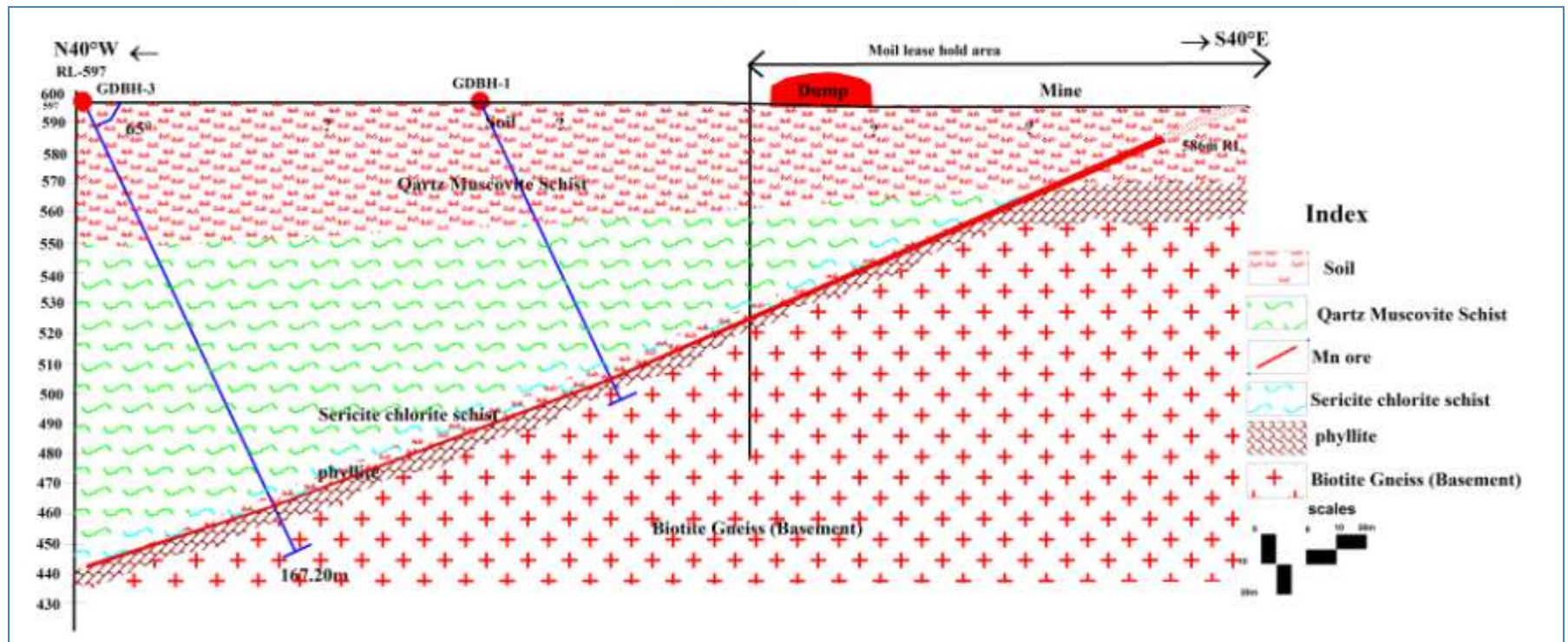


Figure XII.2 Borehole cross section for GDBH-1 & 3.

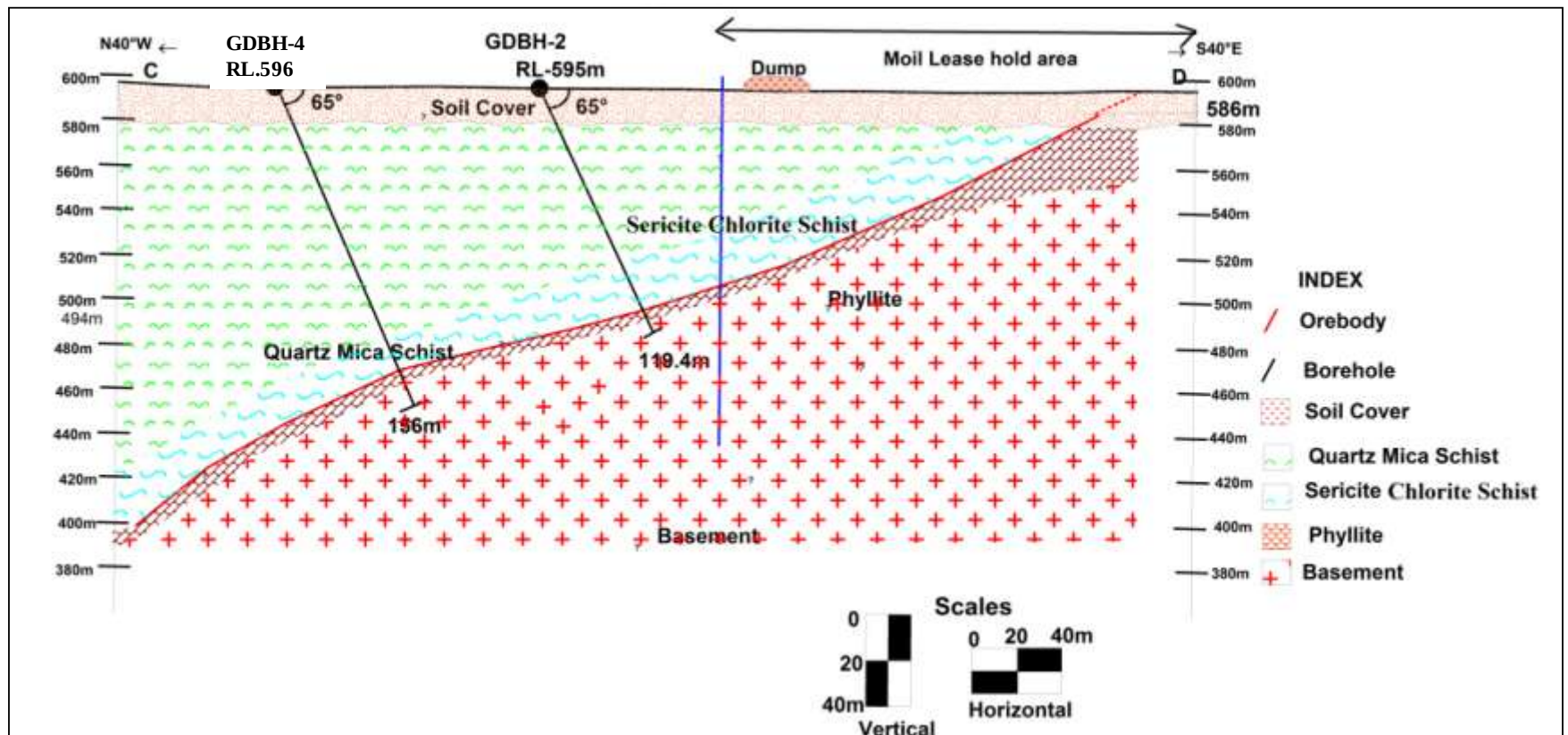


Figure XII.3 Borehole cross section for GDBH-2 & 4.

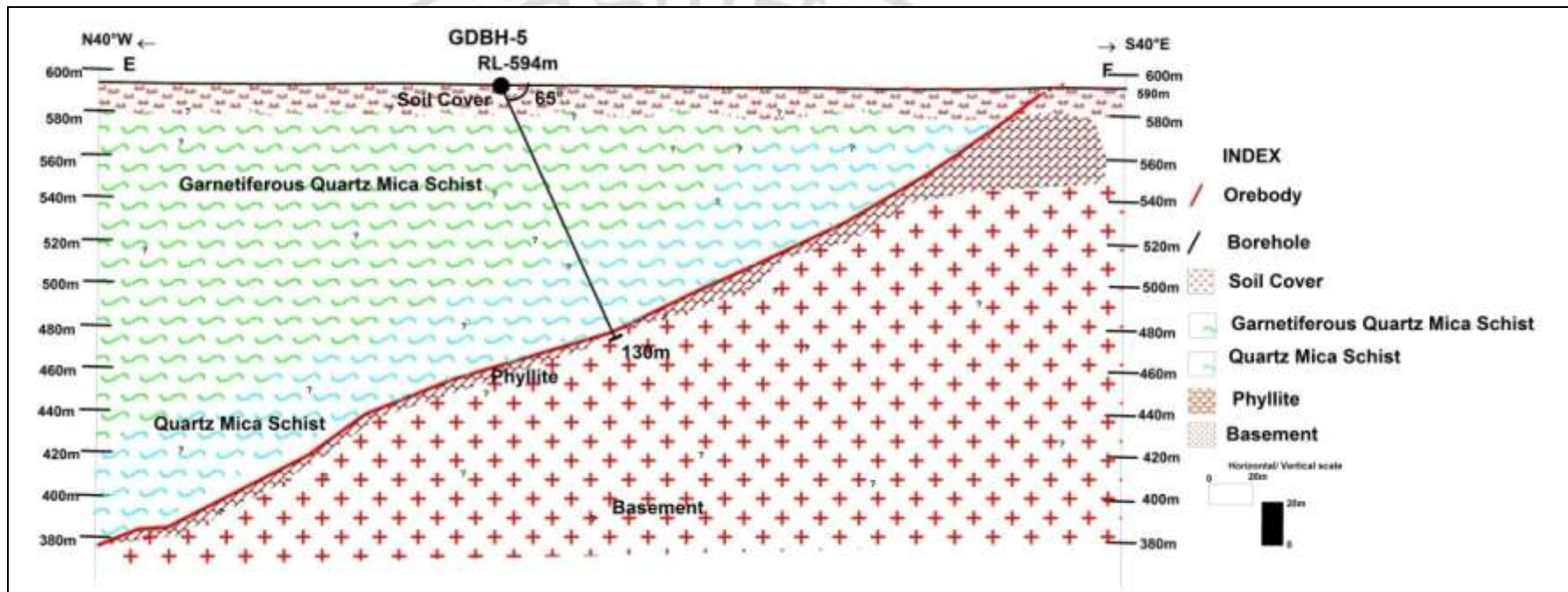


Figure XII.4 Borehole cross section for GDBH-5.

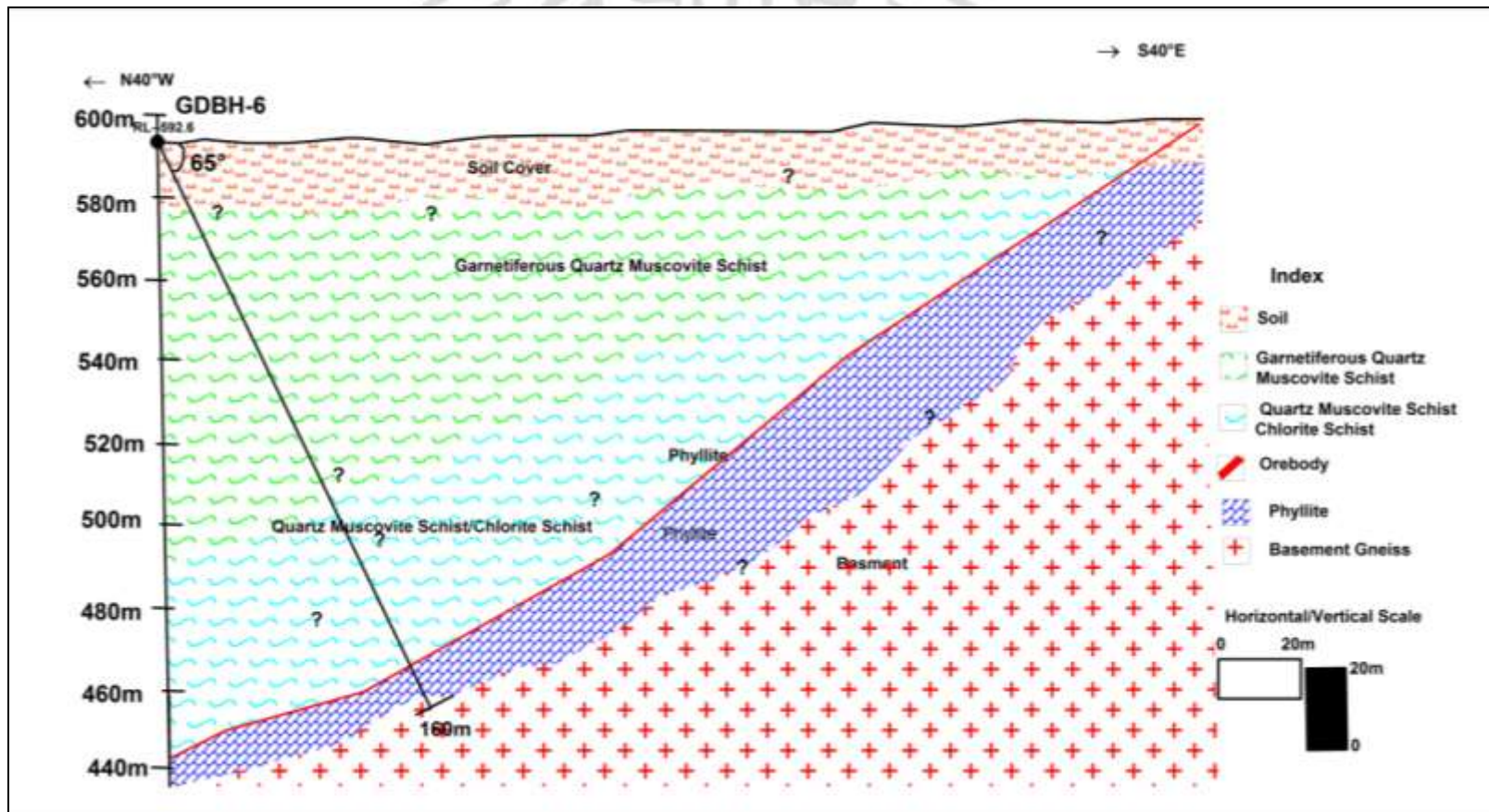


Figure XII.5 Borehole cross section for GDBH-6.

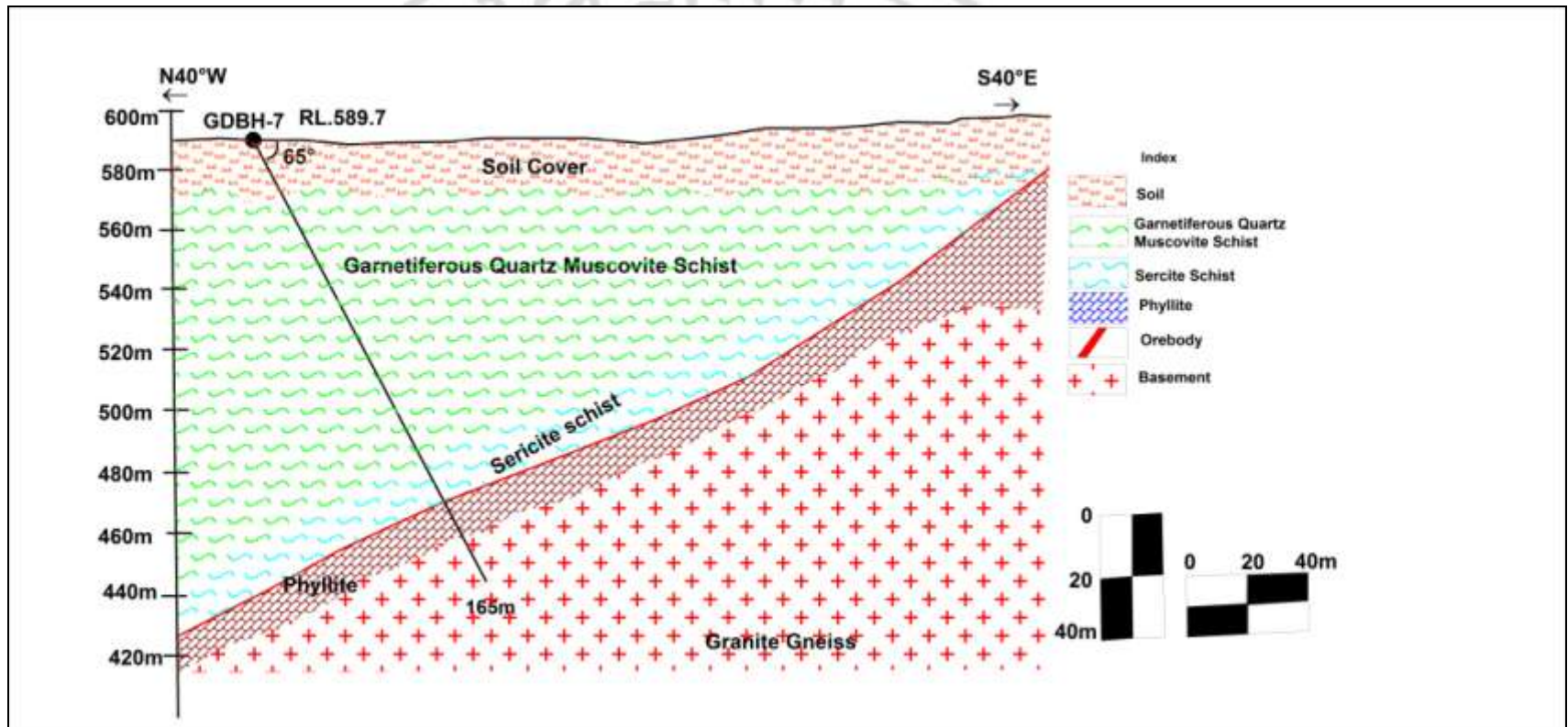


Figure XII.6 Borehole cross section for GDBH-7.

XII.06 Description of lodes

Ore lodes have been described above in 12.2 sections.

XII.07 Preparation of LV sections

In longitudinal vertical section the R.L. of the intersection of the ore body is projected on any vertical plane parallel to the strike of the ore body. Cross sections were prepared across the strike of the ore body for all the section lines, along which boreholes were drilled. L-V section along the strike of the ore body (*Plate-VI*) is also prepared where all the seven boreholes (viz. GDBH-1 to GDBH-7) have intersected ore horizon. Area of influence is taken between half of two boreholes.

XII.08 Preparation of Level Plan

Ore body is not cutting in same level as it is deformed so level Plan is not prepared.

XII.09 Assumptions for resource estimation

The following assumptions were followed while calculating the resource:

1. As the objective was to calculate the resource so all boreholes are taken into consideration drilled during FSP.
2. Area influenced by MOIL boundary is excluded in resource calculation.
3. Strike influence of particular borehole is taken half on either side in LV section. In case of last section lines on either side strike influence is reduced to 25%.
4. The dip length of the intersection by first level of borehole has been considered upto the boundary of the area of MOIL, as the ore body is exposed in the trenches dug by MOIL. The dip influence of first and second level series borehole is taken to be half the distance between two boreholes. 25% of spacing between the boreholes is taken as influence area beyond the second level.
5. For the boreholes with only first level, dip influence is taken as half the depth of intersection along the borehole.
6. For estimates of gross tonnage of material the bulk density used is 2.09 based on beneficiation studies carried out in IBM for an item of FSP 2016-17 of GSI.
7. Tonnage = volume x bulk density (Tonnage is calculated using this formula).

XII.10 Methodology of ore resource estimation

XII.10.1 Longitudinal-Vertical Section method

In LVS method the RL of the intersection of orebody has been projected on vertical plane parallel to the strike of the ore body and lodes have been co-related. The area determined by the LVS has been multiplied by the horizontal thickness of the ore body to know the volume of ore body. i.e $\text{Tonnage} = \text{Horizontal width (cumulative thickness of Mn bands in a borehole measured in cross section)} \times \text{Strike length} \times \text{Dip RL difference (i.e. Difference of higher RL – Lower The intersection R.L. is marked on the appropriate level)}$.

The distance between the vertical lines on either side of the intersection point indicates the effective strike extent for the intersection.

- The horizontal lines placed on above the intersection point and the other below it, indicate the upper and the lower levels respectively between which the dip length is effective.
- The dip of the ore horizon is indicated at the top left hand corner of the panel.
- Tonnage of gross resource(ore + waste) indicated by each intersection is indicated in table 12.3

From LV section method volume of block is calculated 73741.84m^3 and tonnage is 154120.44 ton with average grade of 21.65% (Table-XI).

XII.10.2. Cross section method

There are two rules following which the resource is estimated by Cross Section method.

i) Rule of nearest point.

ii) Rule of Gradual change.

The rule of nearest point of cross sectional area method for estimation of resource has been adopted with reference to the ore zone intersected in each borehole. The entire area along individual cross section is calculated for all the boreholes falling in the same cross section This is also known as rule of equal influence and according to this the value of any point between two stations is considered constant, equal to the value of the nearest station. In a general case of borehole A and B with thickness t_1 and t_2 , the value of each one extent to midpoint 'M' between holes (*Figure XII.7*). Any point on line AB or in the

area of t_1 or t_2 , except M ; is inside the “linear or area of influence” of a station A or B and near to it than to the adjoining one. Thus this property gives the rule its name of nearest point.

Here true thickness is determined by multiplying app thickness of ore horizon (ore + waste) with sin of angle measured in borehole for ore horizon with respect to core axis. By cross section method volume of block is calculated 80849.66m^3 and tonnage is 168975.78 ton with 21.65% grade (Table-X).

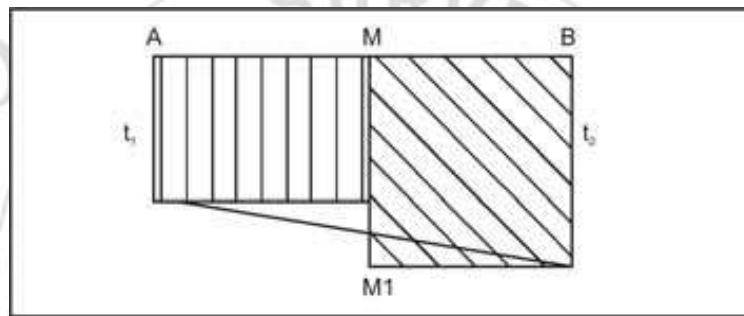


Figure XII.7: Interpretation of values between two boreholes A and B.

Table X Gross resource (ore + waste) of material estimated by cross section method (*Plate-I to V*).

Section Line	B.H. No.	Angle of lode with core axis	App. Thickness along borehole (in m)	True Thickness Calculated (in m)	Dip length (in m)	DIP RL (m)		Strike Length (in m)	Mn (in %)	Bulk Density based on IBM report	Volume M ³	Tonnage
						Upper	Lower					
AB	GDBH-1	85°	1.38	1.25	115	533	485	90	28.07	2.09	12937.5	27039.3
CD	GDBH-2	80°	0.55	0.50	96	538	491	127	30.80	2.09	6096	12740.6
AB	GDBH-3	75°	0.69	0.63	99	485	440	90	19.31	2.09	5613.3	11731.79
CD	GDBH-4	75°	0.24	0.22	73	491	464	127	22.74	2.09	2039.6	4262.8
EF	GDBH-5	80°	1.09	0.99	109.5	525	462	132	16.39	2.09	14309.4	29906.6
GH	GDBH-6	85°	0.60	0.54	168	515	436	174	13.55	2.09	15785.28	32991.23
IJ	GDBH-7	70°	1.18	1.07	138	529	430	163	20.72	2.09	24068.58	50303.33
							Averaged Grade		21.65 %	Total	80849.66 m ³	168975.78 ton (0.168 mt)

Table XI Gross resource of material (ore + waste) estimated by LV section method (*Plate-VI*).

Section Line	B.H. No.	Angle of lode with core axis	App. Thickness (in m)	Horizontal Width as Calculated (in m)	Dip length (m)	Dip RL (m)		Strike Length (m)	Mn (in %)	Bulk Density based on IBM report	Volume M ³	Tonnage
						Upper	Lower					
AB	GDBH-1	85°	1.38	2.749	48	533	485	90	28.07	2.09	11875.68	24820.17
CD	GDBH-2	80°	0.55	0.944	47	538	491	127	30.80	2.09	5634.736	11776.58
AB	GDBH-3	75°	0.69	1.037	45	485	440	90	19.31	2.09	4199.85	8777.68
CD	GDBH-4	75°	0.24	0.361	27	491	464	127	22.74	2.09	1237.86	2587.14
EF	GDBH-5	80°	1.09	1.090	63	525	462	132	16.39	2.09	9064.44	18944.67
GH	GDBH-6	85°	0.60	1.195	79	515	436	174	13.55	2.09	16426.47	34331.32
IJ	GDBH-7	70°	1.18	1.568	99	529	430	163	20.72	2.09	25302.81	52882.88
						Averaged Grade			21.65%	Total	73741.84 m ³	154120.44ton (0.154 mt)

CHAPTER XIII

RECOMMENDATION

After the drilling work it was found that in strike direction ore body shows pinching and swelling. In borehole GDBH-1, 5 & 7 it shows swelling while in GDBH- 2 & 6 ore body shows pinching. In the dip direction along borehole GDBH-1 & 2 ore body shows pinching. In SW side of block ore body is continuing as encountered in borehole GDBH-7. Further exploration may be done in South West direction towards Jagantola which is active mine. Development of this block upto G-2 stage would certainly help to extend the life of existing mine at the later stage through MMDR amendment act, 2015(Auctioning).



CHAPTER XIV

CONCLUSION

Drilling was carried out in Gudma block during FSP 2017-18. Drilling was started on 05.11.2017 and completed on 13.03.2018. A total of 1009.53 m of drilling were done and 7 nos. of boreholes were drilled at second (100m) and third (150m) level to know strike and depth continuity of manganese horizon. A total of 4.38 m thick ore horizon was intersected in 7 different boreholes (GDBH-1, 2,3,4,5,6 & 7) with maximum thickness of ore horizon as 1.38m (GDBH-1) and minimum thickness of ore horizon as 0.24 m (GDBH-4). A total of 44 nos. core samples including 7 nos check samples, 20 nos. BRS, 10 nos. REE, 10 nos. PS. 10 nos. PCS and 5 nos. of XRF samples were collected and submitted to concern lab for analysis. Core Logging along with geophysical logging of two boreholes was done. DGPS survey was carried out and block boundary was demarcated along with borehole points.

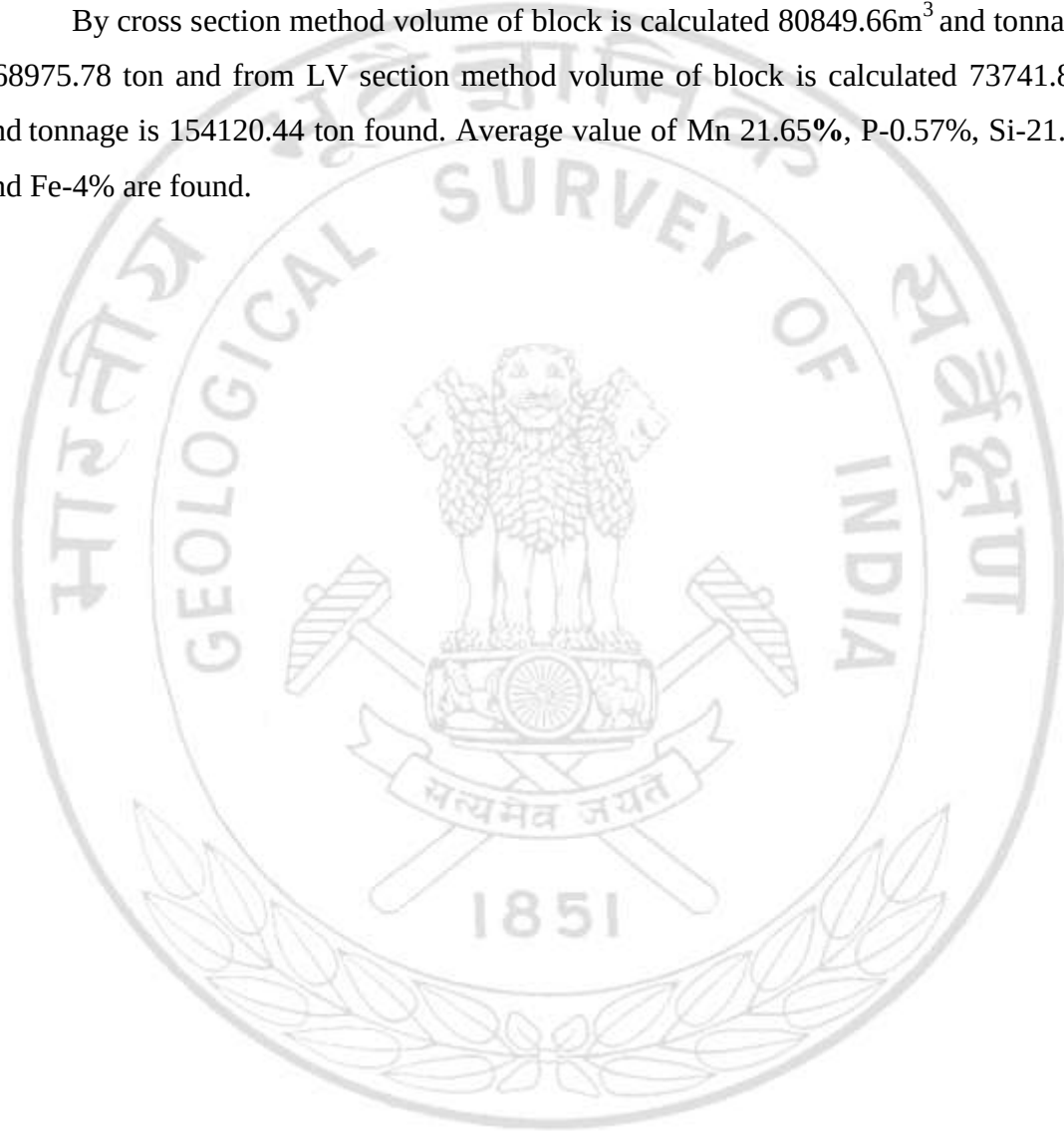
The area exposes rocks of Archaean age in the form of older metamorphics and of Pre-cambrian age as Sausar Group. Both the sequences have a regional strike of ENE-WSW to NE-SW and dip at low to moderate angles towards NW. The area has undergone at least four phases of deformation. The contact between older metamorphics and Sausars is tectonic (thrust after Khan et al 1993-96).

The manganese mineralisation in the area is syn-sedimentary associated with the Sausar Group of rocks. The ore horizon probably transgresses from Lohangi to Mansar formations. Ukwa deposit has been proved to extend over a strike length of 5400 m.

According to earlier workers, the continuity of the ore horizon has been tested for more than 300 m of vertical depth in the eastern part of the deposit and more than 200 m in rest of the deposit over major part of the strike length. In terms of dip length, it measures up to 700 m from the surface exposure of ore horizon in Ukwa mine. Bottoming of ore horizon has not been observed. However, grade and thickness of ore at depth are not very encouraging due to high content of phosphorous at greater depth. Thickness of ore horizon ranges between <15 cm to about 10 m, the average being 3.58 m. (Khan et. al.). The aggregate thickness of the manganese ore bands ranges between 0.23m to 4.85 m, the average being 1.76 m (Kumar et al 2016). Lithologically the ore horizon constitutes a

bed with sharp contacts with its hanging wall and footwall rocks. During current FSP a total of 5.73 ore horizon has been intersected in 7 boreholes. Along the borehole minimum thickness of ore horizon is 0.24m and maximum 1.38m was intersected. Structurally the Gudma ore deposit shows pinch and swell either side of strike. Along dip direction also the ore horizon shows pinching at depth due to culmination and depression.

By cross section method volume of block is calculated 80849.66m^3 and tonnage is 168975.78 ton and from LV section method volume of block is calculated 73741.84m^3 and tonnage is 154120.44 ton found. Average value of Mn 21.65%, P-0.57%, Si-21.32% and Fe-4% are found.



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STATUS OF MINERAL EXPLORATION REPORT IN TERMS OF UNFC COMPLIANCE (2017-18)			
Title of the report	FINAL REPORT ON GENERAL REPORT ON PRELIMINARY EXPLORATION OF MANGANESE MINERALISATION IN GUDMA BLOCK, AN EXTENSION OF WESTERN UKWA BLOCK, BALAGHAT DISTRICT, MADHYA PRADESH (Stage G-3)		
Item no. 065/ME/CR/MP/2017/056	Authors: Amit Kumar, Sr. Geologist. Keshav Khandelwal, Geologist.	Year: 2017-18	
Stage G-2	Nature of compliance required	Complied, Chapter (Page No.)/Not complied	Remarks
I) Summary	(1) Factual Data	C,II (5-10)	
	(2) Recommendations	C, XIII(84)	Objectives was A) To know strike extension of Manganese mineralisation in depth. B) Assessment of manganese ore potential in strike extension of Western Ukwā block.
	(3) Cost estimation	C, II(9)	3,51,560
II) Introduction	(4) Scope & Purpose	C, I(5-10)	
	(5) Authority	NC	
	(6) Current/Post Property Examination)BGI	NC	
	(7) Principal Sources of Information	NC	
	(8) Acknowledgement	NC	
III) Property Description	(9) Size, Continuity, Shape, surface rights	C,III(11-13)	
	(10) Third party issue	NC	
	(11) Climate/rights with dates	C,III(11-13)	
	(12) Land Title Officer	C,III(11-13)	
	(13) Lat/Long/Village/town/district/mine site	C,III(11-13)	Locality Index
	(14) Accessibility/infrastructure	C,III(11-13)	
	(15) Land nature/water/timber/power/gas/agriculture/manpower	NC	
	(16) Environmental issues/plan act/park/preservation	NC	
	(17) Climate/topography	C,III(11-13)	
	(18) Impact on landscape on mining	NC	
	(19) Map of area/Block with roads/accessibility	C	Plate-VIII
IV) History	(20) Previous work highlights	C, IV(14-17)	
	(21) Attachment of previous maps/sections/tables	NC	
	(22) Previous Mining History	NC	
	(23) Reasons for closure	NC	
	(24) Earlier production data	NC	
	(25) Data on nearby mines	NC	
(V) Geoscience investigation			
(1) Geology	(26) Stratigraphy	C, V(20)	
	(27) Lithology	C, V(20-24)	
	(28) Structure (with maps/profiles)	C, V(25-28)	
	(29) Alteration Zones	NC	

	(30) Geotechnical	C,XI(64-70)	
	(31) Adequacy of data density	NC	
	(32) Reliability (quality & quantity)	NC	
	(33) Geological model	NC	
(II) Investigation methods	(34) Instrument & Technique	NC	
	(35) Geology (Map with scale)	C	Plate-VIII
	(36) Geophysical (with maps & scale)	C,V(35-38)	
	(37) Geochemical (with map & scale)	NC	
	(38) Data density of above	NC	
	(39) Size & style of mineralised zone	C, VIII(43-45)	
	(40) Diagrams	NC	
	(41) Metallurgical tests	NC	
	(42) Deleterious contamination source	NC	
(VI) Mineral Deposits			
(I) Features of deposit	(43) Type of mineralisation	C, VIII(43-45)	
	(44) Mode of occurrence	C, VIII(43)	
	(45) Nature of mineralisation (gossan/alteration/structure)	C, VIII(43)	
	(46) Reflectance of Mineralisation on geology/gp/gc	NC	
	(47) Map given	C	Plate-VIII
(II) Sampling details	(48) Quality & Quantity of sample information	NC	
	(49) Reliability on resource estimation	NC	
	(50) Nature of sampling (Bed rock/grab/channel)	NC	
	(51) Sampling method	NC	
	(52) Introduction of duplicates in analysis	NC	
(III) Analysis details	(53) Laboratory	NC	
	(54) Analytical methods	NC	
	(55) Precision & accuracy	NC	
	(56) Cross check analysis details	NC	
(IV) Quality/Grade	(57) Statistical data processing	NC	
	(58) Cut off grade	C, XII(72)	
	(59) Assumptions	C, XII(79)	
(V) Resource estimation	(60) Geological interpretation	C, XII	
	(61) Statement of tonnage & grade	C, XII(80-81)	
	(62) Geometry of mineralisation assumed	C	Fig. XII.1 to XII.6
	(63) Method adopted to arrive estimation	C, XII(79)	
	(64) Estimation	C, 1XII(82-83)	
(VI) Economic viability	(65) Rough estimation of economic viability	NC	
	(66) Intrinsic economic interest	NC	
	(67) Eventual economic exploration	NC	

* Maps and appendix have been incorporated with the report.

CERTIFICATE: Certified that this report is in compliance with UNFC G-3 stage of mineral investigation and resource has been calculated

(AMIT KUMAR)
Geologist, GSI Jabalpur

Place:

Date:

Scrutinizer & Designation:

Signature:

LOCALITY INDEX

Location	Latitudes			Longitudes			
	Degree	Minute	Second	Degree	Minute	Second	Toposheet number.
Aolajhari	21	37	00	79	35	00	64C/01
Baihar	22	03	00	80	33	00	64B/012
Balaghat	21	49	00	80	11	00	64C/01
Baputola	21	53	00	79	44	00	64C/01
Bharweli	21	50	00	80	14	00	64C/01
Chhinditola	21	58	00	80	28	00	64C/05
Ghondi	21	55	00	79	25	00	64C/05
Gudma	21	58	00	80	27	00	64C/05
Jagantola	21	57	00	80	26	00	64C/05
Jagantola	21	57	00	79	16	00	64C/05
Kanhatola	21	52	30	80	21	00	64C/01
Laugur	21	56	00	79	21	00	64C/05
Rupjhar	21	57	00	80	25	00	64C/05
Samnapur	21	58	00	80	29	00	64C/05
Ukwa	21	58	00	80	26	00	64C/05

Analytical results of Bed rock Samples

Annexure-I

Sl. No	Sample No	Lithology	Lat	Long	SiO ₂ (%)	Fe ₂ O ₃ (%)	TiO ₂ (%)	P ₂ O ₅ (%)	MnO (%)	Cu (mg/Kg)	Pb (mg/Kg)	Zn (mg/Kg)	Co(mg/Kg)	Ni (mg/Kg)	Cr (mg/Kg)
1	BRS/01/JBP/Gdm/2017-18	Mn Ore	21.99	80.48	21.54	23.87	0.44	2.60	29.21	410	215	315	95	110	25
2	BRS/02/JBP/Gdm/2017-18	Mn Ore	21.99	80.48	59.14	7.88	0.44	0.91	11.06	170	290	185	40	45	35
3	BRS/03/JBP/Gdm/2017-18	Mn Ore	21.99	80.48	26.37	9.55	0.15	3.83	29.92	740	380	575	85	100	30
4	BRS/04/JBP/Gdm/2017-18	Mn Ore	21.95	80.43	29.23	6.08	0.23	0.26	27.61	240	155	450	45	95	30
5	BRS/05/JBP/Gdm/2017-18	Mn Ore	21.95	80.43	25.26	5.28	0.21	0.25	30.07	200	120	380	30	60	30
6	BRS/06/JBP/GDM/2017-18	Gneiss	21.97	80.45	52.23	4.44	0.72	0.10	0.05	20	40	75	20	15	170
7	BRS/07/JBP/GDM/2017-18	Chlorite Schist	21.97	80.45	61.88	19.15	0.27	0.13	0.16	<10	10	50	35	10	25
8	BRS/08/JBP/GDM/2017-18	Quartz vein with pyrite	21.97	80.45	90.96	0.94	0.05	0.11	0.03	<10	<10	10	100	<10	10
9	BRS/09/JBP/GDM/2017-18	Manganetite bearing sericite schist	21.97	80.45	51.04	4.37	0.92	0.06	0.07	10	20	90	30	15	120
10	BRS/10/JBP/GDM/2017-18	Gneiss	21.97	80.44	56.99	10.31	0.55	0.21	0.05	130	<10	80	30	15	35
11	BRS/11/JBP/GDM/2017-18	Gneiss	21.97	80.45	55.25	6.99	1.07	0.23	0.39	15	30	95	20	10	30
12	BRS/12/JBP/GDM/2017-18	Quartz vein	21.96	80.44	90.21	0.21	0.01	0.03	0.04	10	10	<10	50	10	<10
13	BRS/13/JBP/GDM/2017-18	Garnetiferous quartz muscovite schist	21.95	80.44	53.40	4.70	0.74	0.06	0.16	<10	30	35	20	10	20
14	BRS/14/JBP/GDM/2017-18	Phyllite	21.95	80.44	65.12	4.10	0.61	0.06	0.13	<10	20	25	30	10	20

15	BRS/15/JBP/GDM/2017-18	Pyrite bearing garnetiferous quartz muscovite schist	21.97	80.44	55.26	5.88	1.03	0.07	0.14	70	30	95	35	35	15
16	BRS/16/JBP/GDM/2017-18	Laterite	21.96	80.44	19.07	56.21	2.77	0.31	<0.04	55	20	20	10	15	85
17	BRS/17/JBP/GDM/2017-18	Laterite	21.96	80.46	13.31	23.38	7.89	0.10	0.19	90	15	40	30	35	145
18	BRS/18/JBP/GDM/2017-18	Pyrite bearing garnetiferous quartz muscovite schist	21.96	80.44	50.27	5.27	0.87	0.11	0.09	25	30	90	25	35	80
19	BRS/19/JBP/GDM/2017-18	Laterite	21.96	80.46	21.06	44.02	2.44	0.14	0.04	50	25	25	10	50	145
20	BRS/20/JBP/GDM/2017-18	Manganetitebearing garnetiferous quartz muscovite schist	21.96	80.44	50.40	5.99	1.18	0.11	0.05	15	20	65	35	25	55

Analytical results of REE

Annexure-II

Sl. No.	Sample No	Lithology	Lat	Long	La (mg/Kg)	Ce (mg/Kg)	Pr (mg/Kg)	Nd (mg/Kg)	Eu (mg/Kg)	Sm (mg/Kg)	Gd (mg/Kg)
1	REE/01/JBP/Gdm/2017-18	Quartz vein	21.967	80.44	<1	1.39	0.13	0.53	0.02	0.13	0.09
2	REE/02/JBP/Gdm/2017-18	Quartz vein	21.9673	80.44	9.93	19.44	2.27	8.14	0.33	1.51	1.44
3	REE/03/JBP/Gdm/2017-18	Gneiss	21.9707	80.50	12.12	20.13	2.33	8.48	0.41	1.39	1.21
4	REE/04/JBP/Gdm/2017-18	Granite	21.8954	80.49	56.29	94.45	9.63	30.57	1.07	3.35	3.08
5	REE/05/JBP/Gdm/2017-18	Granite Gneiss	21.9673	80.44	158.86	306.89	34.43	118.04	1.56	15.99	12.51
6	REE/06/JBP/GDM/2017-18	Pyrite bearing Garnetiferous Quartz muscovite Schist	21.9651	80.44	108	216	25	92	2.97	16.73	12.35
7	REE/07/JBP/GDM/2017-18	Soil	21.9651	80.44	39	111	9	35	1.40	6.53	5.93
8	REE/08/JBP/GDM/2017-18	Quartz vein	21.9651	80.44	8	17	2	7	0.24	1.38	1.13
9	REE/09/JBP/GDM/2017-18	Quartz vein	21.9659	80.44	2	3	<0.1	1	0.07	0.33	0.24
10	REE/10/JBP/GDM/2017-18	Quartz vein	21.9659	80.44	<1	<2	<0.1	<0.3	<0.02	<0.1	<0.05
	Sample No	Lithology	Lat	Long	Tb (mg/Kg)	Dy (mg/Kg)	Ho (mg/Kg)	Er (mg/Kg)	Tm (mg/Kg)	Yb (mg/Kg)	Lu (mg/Kg)
1	REE/01/JBP/Gdm/2017-18	Quartz vein	21.967	80.44	0.02	0.09	0.02	0.05	0.01	0.04	0.01
2	REE/02/JBP/Gdm/2017-18	Quartz vein	21.9673	80.44	0.22	1.06	0.21	0.56	0.08	0.54	0.09
3	REE/03/JBP/Gdm/2017-18	Gneiss	21.9707	80.50	0.19	1.06	0.22	0.67	0.10	0.66	0.11
4	REE/04/JBP/Gdm/2017-18	Granite	21.8954	80.49	0.32	1.27	0.22	0.65	0.08	0.48	0.07

5	REE/05/JBP/Gdm/2017-18	Granite Gneiss	21.9673	80.44	1.48	6.44	1.02	2.78	0.33	1.94	0.29
6	REE/06/JBP/GDM/2017-18	Pyrite bearing Garnetiferous Quartz muscovite Schist	21.9651	80.44	1.45	5.41	0.81	2.42	0.32	2.28	0.38
7	REE/07/JBP/GDM/2017-18	Soil	21.9651	80.44	0.99	5.61	1.09	3.73	0.49	3.18	0.49
8	REE/08/JBP/GDM/2017-18	Quartz vein	21.9651	80.44	0.16	0.78	0.14	0.44	0.07	0.44	0.07
9	REE/09/JBP/GDM/2017-18	Quartz vein	21.9659	80.44	0.04	0.22	0.04	0.07	0.02	0.12	0.02
10	REE/10/JBP/GDM/2017-18	Quartz vein	21.9659	80.44	0.01	<0.03	<0.01	<0.02	<0.01	0.02	<0.01



Analytical result of XRF samples

Annexure-III

Sl. No	Sample No	Lithology	Lat	Long	SiO ₂ (%)	Al ₂ O ₃ (%)	Fe ₂ O ₃ (%)	CaO (%)	MgO (%)	Na ₂ O (%)	K ₂ O (%)	TiO ₂ (%)	P ₂ O ₅ (%)
1	XRF/01/JBP/GDM/2017-18	Garnetiferous Quartz muscovite Schist	21.99	80.48	53.28	28.26	6.21	0.10	0.78	0.86	6.73	1.02	0.12
2	XRF/02/JBP/GDM/2017-18	Pyrite bearing quartz vein	21.97	80.44	76.66	10.35	4.80	0.39	0.77	0.35	2.03	0.41	0.34
3	XRF/03/JBP/GDM/2017-18	Phyllite	21.97	80.44	17.72	34.02	19.66	0.17	0.33	0.13	0.13	9.74	0.11
4	XRF/04/JBP/GDM/2017-18	Mn ore	21.97	80.44	27.87	5.42	7.25	0.91	0.09	35.76	0.61	0.24	1.89
5	XRF/05/JBP/GDM/2017-18	Manganiferous Quartzite	21.97	80.44	84.23	1.47	4.98	0.23	<0.1	0.08	0.07	0.09	0.77
Sl. No	Sample No	Lithology	Lat	Long	MnO (%)	LOI (%)	Ba (mg/Kg)	Nb (mg/Kg)	Rb (mg/Kg)	Sr (mg/Kg)	Y (mg/Kg)	Zr (mg/Kg)	
1	XRF/01/JBP/GDM/2017-18	Garnetiferous Quartz muscovite Schist	21.99	80.48	0.02	3.50	1385	17	358	116	51	196	
2	XRF/02/JBP/GDM/2017-18	Pyrite bearing quartz vein	21.97	80.44	0.30	1.66	448	10	102	57	10	73	
3	XRF/03/JBP/GDM/2017-18	Phyllite	21.97	80.44	0.05	15.56	744	36	15	18	13	479	
4	XRF/04/JBP/GDM/2017-18	Mn ore	21.97	80.44	0.38	6.45	31200	<5	38	529	33	32	
5	XRF/05/JBP/GDM/2017-18	Manganiferous Quartzite	21.97	80.44	6.97	1.01	5519	<5	9	317	<10	15	

Analytical results of PCS

Annexure-IV

Sl. No	Sample No	Rock Type	Lat	Long	SiO ₂ (%)	Al ₂ O ₃ (%)	Fe ₂ O ₃ (%)	CaO (%)	MgO (%)	Na ₂ O (%)	K ₂ O (%)	TiO ₂ (%)	P ₂ O ₅ (%)
1	PCS/01/JBP/Gdm/2017-18	Gneiss	21.97	80.44	64.58	14.55	4.50	2.47	1.94	4.08	3.48	0.61	0.27
2	PCS/02/JBP/Gdm/2017-18	Phyllite	21.97	80.44	56.71	19.84	5.59	0.85	2.67	0.20	6.32	0.68	0.12
3	PCS/03/JBP/Gdm/2017-18	Sericite Schist	21.97	80.44	57.63	24.18	5.14	0.31	2.17	0.63	6.27	0.70	0.13
4	PCS/04/JBP/Gdm/2017-18	Garnetiferous Quartz muscovite Schist	21.97	80.44	52.01	28.54	4.57	0.22	1.25	1.26	6.00	1.00	0.10
5	PCS/05/JBP/Gdm/2017-18	Garnetiferous Quartz muscovite Schist	21.99	80.48	51.40	28.83	5.85	0.07	0.71	0.78	7.05	1.03	0.11
6	PCS/06/JBP/Gdm/2017-18	Gneiss	21.97	80.50	75.09	14.29	2.97	0.08	0.25	0.46	3.30	0.49	0.04
7	PCS/07/JBP/GDM/2017-18	Gneiss	21.97	80.45	57.14	22.63	5.50	0.79	2.21	0.81	6.47	0.72	0.12
8	PCS/08/JBP/GDM/2017-18	Magnetite bearing sericite schist	21.97	80.44	33.76	0.98	6.9	0.74	<0.1	0.12	0.75	<0.1	2.3
9	PCS/09/JBP/GDM/2017-18	Pyrite bearing Garnetiferous Quartz muscovite Schist	21.97	80.44	82.69	1.32	4.68	0.21	0.04	0.08	0.23	0.1	0.78
10	PCS/10/JBP/GDM/2017-18	Phyllite	21.97	80.45	52.08	28.86	5.74	0.13	1.12	0.9	6.95	1.1	0.08
Sl. No	Sample No	Rock Type	Lat	Long	MnO (%)	LOI (%)	Ba (mg/Kg)	Nb (mg/Kg)	Rb (mg/Kg)	Sr (mg/Kg)	Y (mg/Kg)	Zr(mg/Kg)	
1	PCS/01/JBP/Gdm/2017-18	Gneiss	21.97	80.44	0.11	2.17	1295	73	204	211	35	439	

2	PCS/02/JBP/Gdm/2017-18	Phyllite	21.97	80.44	3.52	4.33	1542	18	227	362	46	180
3	PCS/03/JBP/Gdm/2017-18	Sericite Schist	21.97	80.44	0.37	3.21	1308	14	293	82	39	141
4	PCS/04/JBP/Gdm/2017-18	Garnetiferous Quartz muscovite Schist	21.97	80.44	0.03	3.10	1146	17	265	57	14	216
5	PCS/05/JBP/Gdm/2017-18	Garnetiferous Quartz muscovite Schist	21.99	80.48	0.02	3.41	1326	16	358	117	44	191
6	PCS/06/JBP/Gdm/2017-18	Gneiss	21.97	80.50	0.02	1.60	166	5	124	38	<10	149
7	PCS/07/JBP/GDM/2017-18	Gneiss	21.97	80.45	0.06	1.40	1009	24	273	60	26	236
8	PCS/08/JBP/GDM/2017-18	Magnetite bearing sericite schist	21.97	80.44	41.98	4.81	17341	<5	22	724	17	15
9	PCS/09/JBP/GDM/2017-18	Pyrite bearing Garnetiferous Quartz muscovite Schist	21.97	80.44	8.81	1.59	5926	<5	14	613	<10	18
10	PCS/10/JBP/GDM/2017-18	Phyllite	21.97	80.45	0.11	3.37	1009	17	367	84	42	220

Analytical result of core samples

Annexure-V

Sr. No.	Sample No	Lat	Long	SiO ₂ (%)	Fe ₂ O ₃ (%)	P ₂ O ₅ (%)	MnO (%)	TiO ₂ (%)
1	CS/01/JBP/GDM/2017-18/GDBH-1	21.9673	80.4491	56.77	4.79	0.87	0.07	0.68
2	CS/02/JBP/GDM/2017-18/GDBH-1	21.9673	80.4491	14.81	8.33	4.26	42.13	0.18
3	CS/03/JBP/GDM/2017-18/GDBH-1	21.9673	80.4491	8.41	7.52	2.86	53.16	0.05
4	CS/04/JBP/GDM/2017-18/GDBH-1	21.9673	80.4491	12.55	3.38	1.14	58.02	0.05
5	CS/05/JBP/GDM/2017-18/GDBH-1	21.9673	80.4491	84.77	2.10	0.61	7.29	0.05
6	CS/06/JBP/GDM/2017-18/GDBH-1	21.9673	80.4491	12.93	5.04	0.95	43.94	0.05
7	CS/07/JBP/GDM/2017-18/GDBH-1	21.9673	80.4491	64.78	7.56	0.21	1.00	0.56
8	CS/01/JBP/GDM/2017-18/GDBH-2	21.967	80.4454	61.73	5.05	0.52	0.10	0.65
9	CS/02/JBP/GDM/2017-18/GDBH-2	21.967	80.4454	22.75	9.57	1.76	45.30	0.18
10	CS/03/JBP/GDM/2017-18/GDBH-2	21.967	80.4454	18.87	6.62	0.97	53.10	0.05
11	CS/04/JBP/GDM/2017-18/GDBH-2	21.967	80.4454	45.52	10.22	1.62	18.92	0.31
12	CS/05/JBP/GDM/2017-18/GDBH-2	21.967	80.4454	58.76	6.41	0.12	0.32	0.71
13	CS/01/JBP/GDM/2017-18/GDBH-3	21.9663	80.4445	64.83	4.37	0.39	0.4	0.55
14	CS/02/JBP/GDM/2017-18/GDBH-3	21.9663	80.4445	84.98	2.75	0.32	7.45	0.03
15	CS/03/JBP/GDM/2017-18/GDBH-3	21.9663	80.4445	20.29	9.3	2.8	40.35	0.1
16	CS/04/JBP/GDM/2017-18/GDBH-3	21.9663	80.4445	27	6.19	0.66	37.20	0.19
17	CS/05/JBP/GDM/2017-18/GDBH-3	21.9663	80.4445	67.54	6.29	0.24	1.61	0.48
18	CS/01/JBP/GDM/2017-18/GDBH-4	21.9673	80.4435	69.74	4.2	0.4	0.09	0.46
19	CS/02/JBP/GDM/2017-18/GDBH-4	21.9673	80.4435	74.24	3.1	0.8	10.43	0.22

20	CS/03/JBP/GDM/2017-18/GDBH-4	21.9673	80.4435	28.13	7.38	0.97	38.82	0.23
21	CS/04/JBP/GDM/2017-18/GDBH-4	21.9673	80.4435	57.84	5.89	0.16	0.28	0.69
22	CS/01/JBP/GDM/2017-18/GDBH-5	21.9659	80.4433	51.97	17.97	1.53	0.39	0.6
23	CS/02/JBP/GDM/2017-18/GDBH-5	21.9659	80.4433	83.27	5.04	0.89	6.64	0.05
24	CS/03/JBP/GDM/2017-18/GDBH-5	21.9659	80.4433	37.49	11.33	1.65	26.11	0.24
25	CS/04/JBP/GDM/2017-18/GDBH-5	21.9659	80.4433	35.05	10.22	1.27	28.24	0.24
26	CS/05/JBP/GDM/2017-18/GDBH-5	21.9659	80.4433	26.18	6.53	0.97	44.44	0.1
27	CS/06/JBP/GDM/2017-18/GDBH-5	21.9659	80.4433	60.51	5.84	0.13	0.12	0.61
28	CS/01/JBP/GDM/2017-18/GDBH-6	21.9651	80.4421	63.4	4.28	0.66	0.11	0.53
29	CS/02/JBP/GDM/2017-18/GDBH-6	21.9651	80.4421	87.96	2.74	0.21	2.99	0.01
30	CS/03/JBP/GDM/2017-18/GDBH-6	21.9651	80.4421	18.27	4.52	4.72	36.47	0.08
31	CS/04/JBP/GDM/2017-18/GDBH-6	21.9651	80.4421	61.28	5.62	0.21	0.07	0.61
32	CS/01/JBP/GDM/2017-18/GDBH-7	21.9638	80.4408	65.33	4.24	0.24	0.04	0.48
33	CS/02/JBP/GDM/2017-18/GDBH-7	21.9638	80.4408	88.25	1.9	0.11	6.26	0.02
34	CS/03/JBP/GDM/2017-18/GDBH-7	21.9638	80.4408	14.25	5.97	0.66	51.65	0.09
35	CS/04/JBP/GDM/2017-18/GDBH-7	21.9638	80.4408	18.53	3.23	0.31	38.80	0.15
36	CS/05/JBP/GDM/2017-18/GDBH-7	21.9638	80.4408	10.26	3.27	0.77	47.12	0.09
37	CS/06/JBP/GDM/2017-18/GDBH-7	21.9638	80.4408	64.16	5.24	0.15	0.06	0.62
38	CHK/01/JBP/GDM/2017-18	21.9673	80.4491	11.05	9.58	4.17	42.89	0.19
39	CHK/02/JBP/GDM/2017-18	21.967	80.4454	17.17	6.25	1.09	54.09	0.08
40	CHK/03/JBP/GDM/2017-18	21.9663	80.4445	16.17	9.42	3.19	40.42	0.1

41	CHK/04/JBP/GDM/2017-18	21.9673	80.4435	75.48	3.35	0.74	11.39	0.21
42	CHK/05/JBP/GDM/2017-18	21.9659	80.4433	88.52	2.89	0.23	3.27	0.01
43	CHK/06/JBP/GDM/2017-18	21.9651	80.4421	32.03	9.52	1.54	29.05	0.25
44	CHK/07/JBP/GDM/2017-18	21.9638	80.4408	11.45	5.59	0.7	51.77	0.09

Sr. No.	Sample No	Lat	Long	Mn%	Si%	Fe%	P%	Ti%
1	CS/01/JBP/GDM/2017-18/GDBH-1	21.9673	80.4491	0.0542	26.4929	3.35017	0.37985	0.40754
2	CS/02/JBP/GDM/2017-18/GDBH-1	21.9673	80.4491	32.636	6.91138	5.82609	1.85996	0.10788
3	CS/03/JBP/GDM/2017-18/GDBH-1	21.9673	80.4491	41.18	3.92469	5.25956	1.2487	0.02997
4	CS/04/JBP/GDM/2017-18/GDBH-1	21.9673	80.4491	44.945	5.85671	2.36401	0.49774	0.02997
5	CS/05/JBP/GDM/2017-18/GDBH-1	21.9673	80.4491	5.6472	39.5596	1.46876	0.26633	0.02997
6	CS/06/JBP/GDM/2017-18/GDBH-1	21.9673	80.4491	34.038	6.03404	3.52503	0.41478	0.02997
7	CS/07/JBP/GDM/2017-18/GDBH-1	21.9673	80.4491	0.7746	30.2309	5.28754	0.09169	0.33562
8	CS/01/JBP/GDM/2017-18/GDBH-2	21.967	80.4454	0.0775	28.8075	3.53202	0.22704	0.38956
9	CS/02/JBP/GDM/2017-18/GDBH-2	21.967	80.4454	35.092	10.6167	6.69335	0.76843	0.10788

10	CS/03/JBP/GDM/2017-18/GDBH-2	21.967	80.4454	41.134	8.80606	4.63009	0.42351	0.02997
11	CS/04/JBP/GDM/2017-18/GDBH-2	21.967	80.4454	14.656	21.2428	7.14797	0.70731	0.18579
12	CS/05/JBP/GDM/2017-18/GDBH-2	21.967	80.4454	0.2479	27.4215	4.48322	0.05239	0.42552
13	CS/01/JBP/GDM/2017-18/GDBH-3	21.9663	80.4445	0.3099	30.2542	3.05642	0.17028	0.32963
14	CS/02/JBP/GDM/2017-18/GDBH-3	21.9663	80.4445	5.7711	39.6576	1.92338	0.13972	0.01798
15	CS/03/JBP/GDM/2017-18/GDBH-3	21.9663	80.4445	31.257	9.46873	6.50451	1.22251	0.05993
16	CS/04/JBP/GDM/2017-18/GDBH-3	21.9663	80.4445	28.817	12.6001	4.32935	0.28816	0.11387
17	CS/05/JBP/GDM/2017-18/GDBH-3	21.9663	80.4445	1.2472	31.5189	4.39929	0.10479	0.28768
18	CS/01/JBP/GDM/2017-18/GDBH-4	21.9673	80.4435	0.0697	32.5456	2.93752	0.17464	0.27569
19	CS/02/JBP/GDM/2017-18/GDBH-4	21.9673	80.4435	8.0796	34.6456	2.16817	0.34929	0.13185
20	CS/03/JBP/GDM/2017-18/GDBH-4	21.9673	80.4435	30.072	13.1274	5.16165	0.42351	0.13785
21	CS/04/JBP/GDM/2017-18/GDBH-4	21.9673	80.4435	0.2169	26.9922	4.11952	0.06986	0.41354
22	CS/01/JBP/GDM/2017-18/GDBH-5	21.9659	80.4433	0.3021	24.2528	12.5684	0.66801	0.3596
23	CS/02/JBP/GDM/2017-18/GDBH-5	21.9659	80.4433	5.1437	38.8596	3.52503	0.38858	0.02997

24	CS/03/JBP/GDM/2017-18/GDBH-5	21.9659	80.4433	20.226	17.4955	7.92432	0.72041	0.14384
25	CS/04/JBP/GDM/2017-18/GDBH-5	21.9659	80.4433	21.876	16.3568	7.14797	0.55449	0.14384
26	CS/05/JBP/GDM/2017-18/GDBH-5	21.9659	80.4433	34.425	12.2174	4.56715	0.42351	0.05993
27	CS/06/JBP/GDM/2017-18/GDBH-5	21.9659	80.4433	0.093	28.2382	4.08455	0.05676	0.36559
28	CS/01/JBP/GDM/2017-18/GDBH-6	21.9651	80.4421	0.0852	29.5869	2.99347	0.28816	0.31764
29	CS/02/JBP/GDM/2017-18/GDBH-6	21.9651	80.4421	2.3162	41.0483	1.91638	0.09169	0.00599
30	CS/03/JBP/GDM/2017-18/GDBH-6	21.9651	80.4421	28.251	8.52606	3.16133	2.0608	0.04795
31	CS/04/JBP/GDM/2017-18/GDBH-6	21.9651	80.4421	0.0542	28.5975	3.93068	0.09169	0.36559
32	CS/01/JBP/GDM/2017-18/GDBH-7	21.9638	80.4408	0.031	30.4876	2.9655	0.10479	0.28768
33	CS/02/JBP/GDM/2017-18/GDBH-7	21.9638	80.4408	4.8493	41.1836	1.32888	0.04803	0.01199
34	CS/03/JBP/GDM/2017-18/GDBH-7	21.9638	80.4408	40.011	6.65005	4.17548	0.28816	0.05394
35	CS/04/JBP/GDM/2017-18/GDBH-7	21.9638	80.4408	30.056	8.6474	2.25909	0.13535	0.0899
36	CS/05/JBP/GDM/2017-18/GDBH-7	21.9638	80.4408	36.501	4.78803	2.28707	0.33619	0.05394
37	CS/06/JBP/GDM/2017-18/GDBH-7	21.9638	80.4408	0.0465	29.9415	3.66491	0.06549	0.37158

38	CHK/01/JBP/GDM/2017-18	21.9673	80.4491	33.225	5.1567	6.70035	1.82066	0.11387
39	CHK/02/JBP/GDM/2017-18	21.967	80.4454	41.901	8.01272	4.37131	0.4759	0.04795
40	CHK/03/JBP/GDM/2017-18	21.9663	80.4445	31.311	7.54605	6.58844	1.39279	0.05993
41	CHK/04/JBP/GDM/2017-18	21.9673	80.4435	8.8232	35.2243	2.34302	0.32309	0.12586
42	CHK/05/JBP/GDM/2017-18	21.9659	80.4433	2.5331	41.3096	2.02129	0.10042	0.00599
43	CHK/06/JBP/GDM/2017-18	21.9651	80.4421	22.503	14.9474	6.65838	0.67238	0.14983
44	CHK/07/JBP/GDM/2017-18	21.9638	80.4408	40.103	5.34337	3.9097	0.30563	0.05394



True width and horizontal width calculation sheet											
Bh. no.	Intersection of MZ in Borehole		Width of MZ in Borehole (m) t'	BH angle α (degree)	dip of MZ δ (degree)	$\theta = 180^\circ - (\alpha + \delta)$	$\sin \theta$	True width (m) $t = t' \sin\{180^\circ - (\alpha + \delta)\}$			Mn assay A (%)
	From	To							$\sin \delta$	Horizontal width	
GDBH-1	99.82	101.2	1.38	65	30	85	0.9961	1.37	0.500	2.749	28.07
GDBH-2	97.45	98.00	0.55	65	35	80	0.9848	0.54	0.574	0.944	30.80
GDBH-3	146.91	147.60	0.69	65	40	75	0.9659	0.67	0.643	1.037	19.31
GDBH-4	128.36	128.60	0.24	65	40	75	0.9659	0.23	0.643	0.361	22.74
GDBH-5	122.81	123.90	1.09	65	80	80	0.9848	1.07	0.985	1.090	16.39
GDBH-6	141.75	142.35	0.60	65	30	85	0.9961	0.60	0.500	1.195	13.55
GDBH-7	139.87	141.05	1.18	65	45	70	0.9396	1.11	0.707	1.568	20.72

Weighted Average Calculation Table

Annexure-VII

Sample No	Lat	Long	Sample	From(in m)	To (in m)	Sample Length in cm	Mn%	Weighted average
CS/02/JBP/GDM/2017-18/GDBH-1	21.9673	80.4491	Mn Ore	99.82	100.269	25	32.63588	28.07247648
CS/03/JBP/GDM/2017-18/GDBH-1	21.9673	80.4491	Mn Ore	100.269	100.575	25	41.18023	
CS/04/JBP/GDM/2017-18/GDBH-1	21.9673	80.4491	Mn Ore	100.575	100.9	25	44.94502	
CS/05/JBP/GDM/2017-18/GDBH-1	21.9673	80.4491	Manganiferous quartzite	99.57	101.2	42.5	5.647177	
CS/06/JBP/GDM/2017-18/GDBH-1	21.9673	80.4491	Mn Ore	100.9	101.2	15	34.03799	
CS/02/JBP/GDM/2017-18/GDBH-2	21.967	80.4454	Mn Ore	97.45	97.77	25	35.09151	30.73289437
CS/03/JBP/GDM/2017-18/GDBH-2	21.967	80.4454	Mn Ore	97.77	97.94	14.1	41.13376	
CS/04/JBP/GDM/2017-18/GDBH-2	21.967	80.4454	Manganiferous quartzite	97.45	98	15.9	14.65632	
CS/02/JBP/GDM/2017-18/GDBH-3	21.9663	80.4445	Manganiferous quartzite	146.96	147.6	31.1	5.77112	19.31368986
CS/03/JBP/GDM/2017-18/GDBH-3	21.9663	80.4445	Mn Ore	146.91	147.427	25	31.25701	
CS/04/JBP/GDM/2017-18/GDBH-3	21.9663	80.4445	Mn Ore	147.427	147.69	12.9	28.81687	
CS/02/JBP/GDM/2017-18/GDBH-4	21.9673	80.4435	Manganiferous quartzite	128.36	128.52	12.5	8.079568	19.07568238
CS/03/JBP/GDM/2017-18/GDBH-4	21.9673	80.4435	Mn Ore	128.4	128.6	12.5	30.0718	

CS/02/JBP/GDM/2017-18/GDBH-5	21.9659	80.4433	Manganiferous quartzite	122.81	123.9	44.4	5.143656	16.39582939
CS/03/JBP/GDM/2017-18/GDBH-5	21.9659	80.4433	Mn Ore	122.86	123.164	25	20.22603	
CS/04/JBP/GDM/2017-18/GDBH-5	21.9659	80.4433	Mn Ore	123.164	123.414	25	21.87603	
CS/05/JBP/GDM/2017-18/GDBH-5	21.9659	80.4433	Mn Ore	123.414	123.58	14.8	34.42531	
CS/02/JBP/GDM/2017-18/GDBH-6	21.9651	80.4421	Manganiferous quartzite	141.8	142.325	34	2.316195	13.554
CS/03/JBP/GDM/2017-18/GDBH-6	21.9651	80.4421	Mn Ore	141.75	142.35	26	28.25138	
CS/02/JBP/GDM/2017-18/GDBH-7	21.9638	80.4408	Manganiferous quartzite	133.87	141.05	56.5	4.84929	20.72390799
CS/03/JBP/GDM/2017-18/GDBH-7	21.9638	80.4408	Mn Ore	139.883	140.305	25	40.01052	
CS/04/JBP/GDM/2017-18/GDBH-7	21.9638	80.4408	Mn Ore	140.305	140.893	25	30.0563	
CS/05/JBP/GDM/2017-18/GDBH-7	21.9638	80.4408	Mn Ore	140.893	141.05	11.5	36.50137	

Determination of weighted average of Mn: Weighted average of Mn of ore in single borehole is determined by following steps.

1. Addition of multiplication of Mn values of ore with their sample length.
2. This value thus determined by total length of ore.

Illustration: In GDBH-4 sample number CS/02/JBP/GDM/2017-18/GDBH-4 and CS/03/JBP/GDM/2017-18/GDBH-4 are considered ore. Therefore their Mn values are multiplied with sample length and values thus obtained are added. ($8 \times 8.07956 + 16 \times 30.0718$). The value thus obtained is divided by total of ore borehole which is 24, which gives the value 22.74.